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# Gluing Local Contexts into Global Meaning: A Sheaf-Theoretic Decomposition of Transformer Representations

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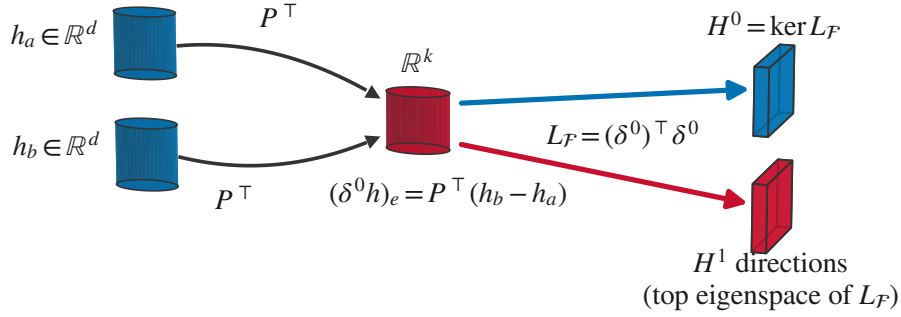


Figure 1: Sheaf decomposition of a paraphrase pair. Hidden states  $h_a, h_b$  pass through  $P^\top$  into the edge stalk  $\mathbb{R}^k$ , where the coboundary measures their disagreement,  $(\delta^0 h)_e = P^\top(h_b - h_a)$ . The bottom and top eigenspaces of the sheaf Laplacian  $L_F$  split the edge stalk  $\mathbb{R}^k$  into  $H^0$  (blue, content stable across phrasing) and the  $H^1$  directions (red, context-dependent variation).

## Abstract

Existing interpretability tools do not test whether a representation keeps the same content when only the phrasing changes. We treat each paraphrase pair as a local witness that two activations should agree and build a cellular sheaf over the paraphrase graph; its Laplacian  $L_F$  answers the question on a frozen model with no training. The kernel  $H^0$  is a content subspace: at 20 dimensions it beats the preserved subspace of the concept eraser LEACE on held-out CounterFact fact retrieval across eight architectures by a mean +17.1 pp, and constraining all paraphrases of a fact jointly adds a further +6.5 pp. The spectral complement concentrates causal influence: ablating the most paraphrase-variable coordinates, a coordinate-wise proxy for it, changes model outputs 5.6 to  $26.5\times$  more than ablating coordinates of matched per-coordinate variability, across nine models from 124M to 13B parameters. Stacking layers into a grid creates  $b_1 = N(L - 1)$  algebraic 1-cycles, and the resulting model-derived harmonic mass varies by nearly three orders of magnitude across six architectures and ranks them as their steering fragility does (Spearman  $\rho = 0.90$ , exact one-sided permutation  $p = 0.014$ ). The same paraphrase graph yields the content subspace, the influence directions, and the fragility measure. Project page: <https://cwru-aism.github.io/gluing-lc-page/>.

# 1 Introduction

Transformer representations entangle what a sentence means with how it is phrased. The hidden state for “The Eiffel Tower is in Paris” differs from “Paris is home to the Eiffel Tower” despite identical semantic content. Existing tools decompose one activation into features [7, 12], test the linear decodability of attributes already chosen [4], or relate two activation spaces to each other [40, 52]. None of them tests cross-context consistency, meaning whether a feature encodes the same information when the phrasing changes and the meaning does not. The gap shows up in steering, where methods improve some models and fail or degrade others [14], and even within one model steerability varies widely across inputs [57]; no existing tool predicts which regime a model is in. We treat the problem as local-to-global agreement. Each paraphrase pair is a local witness that two activations should agree, and together the witnesses should determine one global answer.

In the cellular sheaf we build on the paraphrase graph, hidden states sit on the nodes, edges connect paraphrase pairs, and a learned restriction map  $P^\top$  ( $P \in \mathbb{R}^{d \times k}$ ) projects each pair into a shared  $k$ -dimensional comparison space. The coboundary  $\delta^0$  measures pairwise disagreement, and the sheaf Laplacian  $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$  has a kernel  $H^0 = \ker(\delta^0)$  of globally consistent sections together with a spectral complement of high-disagreement directions. We call the spectral complement the “ $H^1$  directions” (Remark 3.3 explains the name). Section 3 defines the construction, and Section 5 tests it on nine models.

A single-layer matching graph has no cycles, so algebraic  $H^1 = 0$  and the construction reduces to projected within-pair covariance (Proposition 3.1). Stacking the same activations across layers creates cycles. The grid has one node per (sentence, layer) pair, with horizontal edges between paraphrases in the same layer and vertical edges between adjacent layers along each sentence. It carries  $b_1 = N(L - 1)$  algebraic 1-cycles ( $b_1$ , the first Betti number, counts the independent cycles in the graph), one per elementary square, and each square asks whether the layer- $l$ -to-layer- $l+1$  transformation commutes with paraphrase equivalence. The model-derived Hodge harmonic mass (the fraction of a model-derived edge cochain’s energy in the harmonic subspace, the part supported on cycles that is neither a gradient nor a curl) at  $L = 2$  varies by nearly three orders of magnitude across six architectures ( $10^{-4}$  to 0.08; Table 1) against a 0.25 topological reference, and Section 5.4 tests it as a diagnostic of steering fragility. Richer paraphrase graphs also improve  $H^0$ : constraining all paraphrases of a fact jointly beats pair-only  $H^0$  by +6.5 pp on all eight architectures, and an edge-count-matched control attributes the gain to the joint constraint, with the cycle itself retrieval-neutral (Section 5.2). The coboundary of a cellular sheaf on a paraphrase graph yields  $L_{\mathcal{F}}$ , whose kernel is the content subspace (Section 5.2), whose spectral complement concentrates functional influence (Section 5.3), and whose harmonic mass on the depth-induced cycles is tested as a diagnostic of steering fragility (Section 5.4).

Our contributions are:

1. **Content from the kernel.** Sheaf  $H^0$  at 20 dimensions beats LEACE’s preserved subspace at full hidden dimension on held-out CounterFact retrieval across eight architectures (mean +17.1 pp; 85.3% vs. 43.1% on GPT-2). Random or identity restriction maps keep most of the gain, and constraining all of a fact’s paraphrases jointly adds +6.5 pp (Section 5.2).
2. **Influence from the spectral complement.** Ablating the most paraphrase-variable coordinates, a coordinate-wise proxy for the top eigenspace of  $L_{\mathcal{F}}$ , changes model outputs 5.6 to 26.5 $\times$  more than ablating coordinates of matched per-coordinate variability on nine architectures from 124M to 13B parameters. Projecting out the sheaf  $H^1$  eigenbasis shifts next-token distributions 17.8–398 $\times$  more than random subspaces and collapses fact recall to 0–5%, while matched random ablation leaves recall at baseline on the 7B models (Section 5.3, Appendices B.6 and B.7).
3. **Fragility from harmonic mass.** Multi-layer grids carry  $b_1 = N(L - 1)$  algebraic 1-cycles and a model-derived harmonic mass spanning  $[10^{-4}, 0.08]$  that ranks six architectures by steering fragility ( $\rho = 0.90$ ,  $p = 0.014$ ). On Llama-2-7B, contrastive sheaf steering preserves 31.0% of facts against 4.2% for random steering (Section 5.4).

## 2 Cellular Sheaves on Paraphrase Graphs

A cellular sheaf  $\mathcal{F}$  on a graph  $G = (V, E)$  assigns a vector space (stalk)  $\mathcal{F}_v = \mathbb{R}^{d_v}$  to each node and a stalk  $\mathcal{F}_e = \mathbb{R}^{d_e}$  to each edge, connected by linear restriction maps  $\mathcal{F}_{v \triangleleft e} : \mathcal{F}_v \rightarrow \mathcal{F}_e$  at node-edge incidences [13, 31]. The coboundary operator  $\delta^0 : C^0(\mathcal{F}) \rightarrow C^1(\mathcal{F})$  measures local disagreement:

$$(\delta^0 x)_e = \mathcal{F}_{v \triangleleft e}(x_v) - \mathcal{F}_{u \triangleleft e}(x_u) \quad (1)$$

for edge  $e = (u, v)$ . When  $\delta^0 x = 0$  everywhere, local data is *globally consistent*: it glues into a global section. The sheaf Laplacian  $L_{\mathcal{F}} = (\delta^0)^\top \delta^0$  generalizes the graph Laplacian [31]; Figure 1 sketches the pipeline from node stalks through  $\delta^0$  to the  $H^0/H^1$  split. Because  $x^\top L_{\mathcal{F}} x = \|\delta^0 x\|^2$ ,  $\ker(L_{\mathcal{F}}) = \ker(\delta^0) = H^0(\mathcal{F})$  [31], the space of globally consistent sections. The first cohomology  $H^1 = \ker(\delta^1)/\text{im}(\delta^0)$  captures obstructions to gluing. On a graph  $\delta^1 = 0$ , so  $H^1 = C^1/\text{im}(\delta^0)$ : edge assignments that are not the coboundary of any node assignment.

We read  $H^0$  as content that is consistent across semantically equivalent contexts and the spectral complement of  $L_{\mathcal{F}}$  as context-dependent variation: syntax, phrasing, surface form [18, 7]. This reading comes from the definition of the kernel. A direction on which every paraphrase pair of a fact agrees keeps its value when the phrasing varies and the meaning holds, so it tracks the fact’s content, and the kernel is the span of such directions. Cycles indicate where steering may fail. A cycle around which the model-derived cochain fails to sum to zero (a nonzero harmonic component) is a place where the model’s layer-to-layer maps stop commuting with paraphrase equivalence. A fact then travels along inconsistent representational paths, and a direction read off one path can diverge along the others. On a matching graph (one edge per paraphrase pair) the graph is acyclic, so  $b_1(G) = 0$ ,  $\delta^0$  is surjective onto  $C^1$  whenever  $P$  has full column rank  $k$ , and algebraic  $H^1 = \ker(\delta^1)/\text{im}(\delta^0) = 0$ . For transformers we put hidden states on the nodes, paraphrase pairs on the edges, and a restriction map  $P^\top$  that projects from the full activation space to a  $k$ -dimensional edge space. The spectral decomposition lives in edge space, and its eigenvectors pull back to the activation space via  $P$ . The choice of  $P$  (PCA, canonical correlation analysis (CCA), or a contrastive Fisher discriminant) is the only knob, and it changes the decomposition.

LEACE (least-squares concept erasure) [5] fits a concept eraser, well suited to removing the paraphrase / non-paraphrase distinction; the complementary target for content stability is the subspace of minimal within-pair variation. Sparse autoencoders [7, 12] factor a single activation into features, and centered kernel alignment (CKA) and SVCCA [40, 52] score the similarity of two whole spaces; cross-context consistency asks each space to be split into a part that is invariant across phrasing and a part that varies with it. With that split, an intervention can move the variable directions and leave content intact. The coboundary produces the split directly, and Section 5 uses the resulting subspaces for its interventions.

## 3 Local Construction on Matching Graphs

**Sheaf.** The interpretability sheaf  $\mathcal{F}$  over a context graph  $G = (V, E)$  assigns node stalks  $\mathcal{F}(v) = \mathbb{R}^d$ , in which the hidden states of one context form a 0-cochain, edge stalks  $\mathcal{F}(e) = \mathbb{R}^k$  ( $k \ll d$ ) as a shared comparison space, and linear restriction maps  $\mathcal{F}_{v \triangleleft e} = P^\top$ . The matching graph ( $|V| = 2N$ ,  $|E| = N$ , no higher simplices) is the simplest case. We learn  $P \in \mathbb{R}^{d \times k}$  via three concrete choices: joint PCA (top  $k$  components of paired activations), CCA (solving  $C_{12}C_{22}^{-1}C_{21}w = \lambda C_{11}w$  for the top  $k$  eigenvectors), or contrastive Fisher. The contrastive Fisher map solves  $C_W w = \lambda C_T w$  for  $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$  and  $C_T = \frac{1}{N} \sum_i h_i h_i^\top$ , where  $\Delta_i = h_i^{(1)} - h_i^{(2)}$  is the activation difference of pair  $i$ , and takes the  $k$  largest generalized eigenvectors as columns of  $P$  (sign opposite to LDA, orienting  $P$  toward maximal within-pair variance relative to total).  $L_{\mathcal{F}} = P^\top C_W P$  then has maximal spread, the spectral  $H^1$  complement is more selectively concentrated, and  $H^0$  picks up residual content within the highly variable basis.

**Coboundary, Laplacian, decomposition.** Orienting each edge as  $e = (v_2, v_1)$ , with  $h^{(1)}$  at  $v_1$ , the coboundary  $(\delta^0 s)_e = P^\top s_{v_1} - P^\top s_{v_2}$  measures pair disagreement, and the consistency energy  $E_{\text{cons}}(s) = \sum_e \|(\delta^0 s)_e\|^2$  vanishes on  $H^0 = \ker(\delta^0)$ . The empirical sheaf Laplacian in edge space is

$$L_{\mathcal{F}} = \frac{1}{N} \sum_{i=1}^N \Delta_i^{(k)} (\Delta_i^{(k)})^\top \in \mathbb{R}^{k \times k}, \quad \Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)}. \quad (2)$$

This  $k \times k$  matrix equals  $\frac{1}{N} Y L_G Y^\top$  with  $Y = [P^\top h_v]_v$  and  $L_G$  the Laplacian of the matching graph. It is the second-moment matrix of the edge values of  $\delta^0 h$ , not the  $|V|d \times |V|d$  operator  $(\delta^0)^\top \delta^0$ , and we keep the symbol  $L_{\mathcal{F}}$  for it. For a readout  $w \in \mathbb{R}^k$ ,  $w^\top L_{\mathcal{F}} w$  is the Laplacian quadratic form of the observed data seen through  $w^\top P^\top$  [30, 21], and minimising it is a locality preserving projection on the paraphrase graph [32]. Its kernel is the set of readouts under which the data is a global section:  $v \mapsto w^\top P^\top h_v$  is constant along every edge, equivalently  $w^\top \Delta_i^{(k)} = 0$  for all  $i$ . We define  $H^0$  as the span of the bottom  $m$  eigenvectors of  $L_{\mathcal{F}}$  (the low-energy approximation to  $\ker L_{\mathcal{F}}$ ) and  $H^1$  directions as the span of the top  $m$ . The hidden-space pullback is  $\tilde{v}_j = P v_j$ , and the  $H^0$  projector is  $\Pi_{H^0} = P V_{H^0} V_{H^0}^\top P^\top$  ( $P$  has orthonormal columns for the joint-PCA map, so  $\Pi_{H^0}$  is an orthogonal projector; the CCA and contrastive maps of Section 5.4 have unit-norm columns that are not orthogonalised). The eigenproblem is  $k \times k$  ( $k = 128$ ) whatever the hidden dimension, so the spectral step costs the same from 124M to 13B models.

**Steering.** Given a style direction  $\Delta = \bar{h}_{\text{target}} - \bar{h}_{\text{source}}$ , we form (a)  $\Delta_{H^1} = \Delta - \Pi_{H^0} \Delta$  ( $H^0$ -removed) or (b) the projection of  $\Delta$  onto the top  $m$   $H^1$  dimensions weighted by  $s_j = \sum_\ell (\tilde{v}_\ell)_j^2 \lambda_\ell$ . All steering vectors are norm-matched to  $\alpha = 0.3 \|\bar{h}\|$ . Baselines: random, full style (contrastive activation addition, CAA [53]), mean-of-differences, variance top- $k$ , Fisher LDA.

**The matching-graph limitation.** The construction so far gives local consistency:  $\delta^0$  measures within-pair disagreement and  $H^0$  contains its kernel. The graph itself, however, has no cycles, and  $P^\top$  is surjective ( $P$  has full column rank), so the cohomology vanishes above degree zero.

**Proposition 3.1** (Reduction to projected within-pair covariance). *With  $C_W = \frac{1}{N} \sum_i \Delta_i \Delta_i^\top$  and  $C_T = \frac{1}{N} \sum_i h_i h_i^\top$ , on a matching graph with one  $P$  shared by every node and identity transports on edges: (i)  $L_{\mathcal{F}} = P^\top C_W P$ ; (ii) the Fisher baseline is the generalized eigenproblem  $L_{\mathcal{F}} v = \lambda (P^\top C_T P) v$ ; (iii)  $\lambda = \frac{1}{N} \sum_i (\Delta_i^\top P v)^2$  for a unit eigenvector  $v$ . Proof: Appendix A.5.*

**Remark 3.2** (Per-node restriction maps). With per-node maps  $P_v$  and a Procrustes transport  $Q_{uv}$  (the orthogonal map that best aligns stalk  $u$  to stalk  $v$ ), the coboundary is  $(\delta^0 x)_e = P_v^\top x_v - Q_{uv} P_u^\top x_u$ , and  $L_{\mathcal{F}}$  is in general not  $P^\top C_W P$ , nor  $P^\top M P$  for any  $M$  built from hidden-state second moments alone, because the cross-terms  $-P_v^\top h_v h_u^\top P_u Q_{uv}^\top$  carry different maps on the two sides.

**Remark 3.3** (Terminology:  $H^1$  directions). Matching graphs are acyclic, so algebraic  $H^1 = \ker(\delta^1)/\text{im}(\delta^0) = 0$ . We call the top eigenspace of  $L_{\mathcal{F}}$  the  $H^1$  directions as a mnemonic for high disagreement; it is not a cohomology class. On a matching graph every 1-cochain,  $\delta^0 h$  included, is a coboundary, so the name refers to where  $\delta^0 h$  carries the most energy. Algebraic  $H^1 \neq 0$  appears only on the cycle-bearing topologies of Section 4.

Proposition 3.1 needs a single  $P$  shared across nodes with identity transports. It holds on any graph, and we apply it on the matching graph. Within that regime, the restriction-map ablation (Table 11) places the content in the kernel of the within-pair difference covariance whatever  $P$  is, since random or identity maps recover most of the  $H^0$  retrieval; empirically the Fisher baseline of part (ii) degrades fact preservation (15.4% vs. 29.8%). Outside it, multi-layer grids ( $b_1 = N(L-1)$ ) carry a model-derived harmonic mass (Section 4.1), and per-node  $P_v$  with Procrustes transports break the reduction (Remark 3.2, Appendix B.12).

## 4 Multi-Layer Matching Grids and Algebraic $H^1$

A single-layer matching graph trivializes algebraic  $H^1$  because it has no cycles and  $P^\top$  is surjective. Filling paraphrase cliques does not fix this: each filled clique is simply connected, so  $H^1$  vanishes (its  $H^2$  does not). The most direct source of cycles is network depth.

### 4.1 Multi-layer matching grids

Let the vertex set be  $V = \{(s, l) : s \in \text{sentences}, l \in \text{layers}\}$ . Horizontal edges connect paraphrase pairs at each layer, and vertical edges connect each sentence at adjacent layers. The resulting graph (a 1-dimensional complex) is a grid: for  $N$  pairs and  $L$  layers it has  $|V| = 2NL$ ,  $|E| = NL + 2N(L-1)$ , and no 2-faces. Direct computation gives  $b_1 = N(L-1)$  algebraic 1-cycles. Each elementary cycle is a square (sentence  $s$ , paraphrase  $s'$ ) at layers  $l$  and  $l+1$ : the cycle asks whether the layer- $l \rightarrow l+1$  transformation commutes with paraphrase equivalence. When it does,  $\omega$  sums to zero around the

Table 1: Multi-layer matching grid,  $b_1 = N(L-1)$  from depth alone. Hodge harmonic mass of model-derived edge cochains varies by nearly three orders of magnitude. Every row shares the same graph ( $b_1 = 30$  at  $L = 2$ ), so the variation comes from the model’s layer transitions. 30 MRPC pairs with joint-PCA per layer at shared  $k = 8$ . The last column is steering fragility: CounterFact fact preservation (%) under random steering (Section 5.4, Appendix A.3).

| Architecture | $L = 2$ | $L = 4$ | $L = 8$ | $L = 16$ | Pres. (%) |
|--------------|---------|---------|---------|----------|-----------|
| Mistral-7B   | 0.0001  | 0.0001  | 0.0004  | 0.0008   | 39.4      |
| Qwen-2.5-7B  | 0.0001  | 0.018   | 0.016   | 0.096    | 34.3      |
| Llama-3-8B   | 0.001   | 0.002   | 0.003   | 0.004    | 36.0      |
| Phi-2        | 0.033   | 0.126   | 0.109   | 0.067    | 27.5      |
| GPT-2 (124M) | 0.036   | 0.107   | 0.083   | –        | 9.1       |
| Llama-2-7B   | 0.079   | 0.071   | 0.047   | 0.028    | 4.2       |

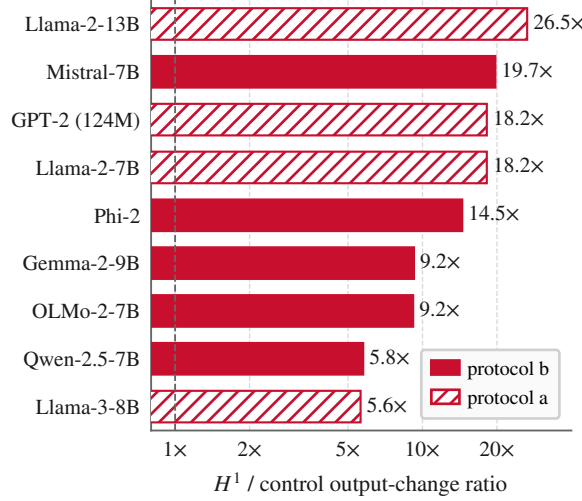


Figure 2: Per-architecture ratio of the output change from ablating the  $H^1$  coordinates to that from ablating control coordinates of matched per-coordinate variability (Table 3). Hatched bars use protocol *a* (one layer, 64 coordinates, effect on logits) and solid bars protocol *b* (three depths, 20 coordinates, effect on the final hidden state). Every architecture exceeds the dashed unit baseline, by  $5.6\times$  to  $26.5\times$ .

square, so  $\omega$  restricted to that cycle is exact and has no component on the harmonic 1-cochain dual to it; when it does not,  $\omega$  has a nonzero harmonic component along that cycle.

**Definition of the model-derived edge cochain  $\omega$ .** Each layer  $l$  has its own joint-PCA restriction map  $P_l^\top$  ( $P_l \in \mathbb{R}^{d \times k}$ ), shared across sentences, and  $h_s^l$  is the mean-pooled hidden state of sentence  $s$  at layer  $l$ . A horizontal edge between paraphrases  $a, b$  at layer  $l$  carries  $\omega(e) = P_l^\top (h_b^l - h_a^l)$ . A vertical edge for sentence  $s$  from layer  $l$  to  $l+1$  carries  $\omega(e) = (P_{l+1}^\top h_s^{l+1} - \mu_{l+1}) - R_{l \rightarrow l+1}^\top (P_l^\top h_s^l - \mu_l)$ , where  $\mu_l$  is the mean projected activation at layer  $l$  and  $R_{l \rightarrow l+1}$  is the orthogonal Procrustes map between the centred projected activations of all  $2N$  sentences at the two layers. With  $R = I$  (centring is immaterial)  $\omega$  would be the coboundary of a node potential and its harmonic mass would be zero; the fitted transport is what makes the mass model-dependent. Stacking gives  $\omega \in \mathbb{R}^{|E| \times k}$ , and the Hodge harmonic mass is  $\|\Pi_{\ker L_1} \omega\|^2 / \|\omega\|^2$  with  $L_1 = \delta^0 (\delta^0)^\top + (\delta^1)^\top \delta^1$  the Hodge 1-Laplacian, which reduces to the edge Laplacian  $\delta^0 (\delta^0)^\top$  on the grid since it has no 2-cells. With all edge stalks identified with one  $\mathbb{R}^k$ , this is the Hodge decomposition of the constant sheaf  $\mathbb{R}^k$  applied to  $\omega$  coordinate-wise [31, 54]. Since  $\omega$  is the coboundary of the potentials  $P_l^\top h_s^l - \mu_l$  under the transport  $R_{l \rightarrow l+1}$ , its harmonic part comes only from  $I - R_{l \rightarrow l+1}^\top$  acting on within-pair differences, and the mass is computed in the joint-PCA frames  $P_l$ . The Hodge decomposition  $C^1 = \text{im } \delta^0 \oplus \ker L_1 \oplus \text{im } (\delta^1)^\top$  is orthogonal [43]. Sensitivity to  $k$  and to alternative restriction maps is reported in Appendix A.2 (the span and the split between robust and fragile architectures are preserved at  $k \in \{8, 64, 128\}$ ).

The Betti number  $b_1$  scales linearly in  $L$  from zero, so depth is a single knob that controls the cycle structure, and the model-derived harmonic mass spans  $[10^{-4}, 0.08]$  at  $L = 2$  (Table 1). A random Gaussian cochain has expected harmonic fraction exactly  $b_1/|E| = 0.25$  at  $L = 2$ ; that value is the random-probe reference. The model’s  $\omega$  places near-zero weight (about  $10^{-4}$ ) on the harmonic subspace for Mistral-7B and Qwen-2.5-7B and substantial weight for Llama-2-7B. Scaling  $k$  from 8 to 128 and triangulating each square preserve the split between the three low-mass and three high-mass architectures (Appendix A.2). Section 5.4 tests whether the mass predicts steering fragility.

**Why the 2-simplices stay unfilled.** The grid has no 2-cells, so its cohomology vanishes above degree 1 and, with constant  $\mathbb{R}$  coefficients,  $\dim H^1 = b_1 = N(L - 1)$  ( $k b_1$  with  $\mathbb{R}^k$  stalks). Filling each square with two 2-simplices sharing a diagonal  $(a, l)$ - $(b, l + 1)$  makes the square cycle the boundary of  $T_1 - T_2$  (with vertex-order orientations); direct computation on that 2-complex gives  $b_1 = b_2 = 0$  for every  $L$  and  $N$  tested, so the harmonic mass is identically zero. What survives is the coexact fraction of  $\omega$  (the curl of  $\omega$  around each filled triangle), which preserves the split between architectures (Appendix A.2). The holonomy of a cycle  $\gamma$  is the composite  $H_\gamma$  of the orthogonal transports along its edges, and the obstruction satisfies  $\|I - H_\gamma\|_F^2 = 2k - 2 \operatorname{tr} H_\gamma \leq 4k$ , so it is at most  $2\sqrt{k}$ , and a Haar-random orthogonal matrix gives  $\mathbb{E}\|I - H_\gamma\|_F^2 = 2k$  (RMS value  $\sqrt{2k}$ ); raw holonomy magnitudes therefore need the null calibration of Appendix B.12.

## 4.2 Relation hypergraphs

A second construction yields cycles from relation structure rather than depth. Let  $\mathcal{D}$  contain  $n_f$  facts with  $n_p$  paraphrases each, partitioned into  $n_r$  relations. For each fact we add a ring on its  $n_p$  paraphrase nodes, and for each relation a ring through the first paraphrase of each of its facts, leaving 2-simplices unfilled. With  $n_f = 60$ ,  $n_p = 5$ ,  $n_r = 6$  this gives 300 nodes,  $300 + 60 = 360$  edges and no faces, with  $\dim H^0 = 6$  and  $\dim H^1 = 66$  (Euler  $\chi = -60 = 6 - 66$ ). The 66 classes decompose as 60 within-fact paraphrase rings (one per fact) plus 6 relation-level cycles (one per relation). The matching graph on the same nodes has  $\dim H^1 = 0$ , and filling fact cliques into 2-simplices also gives  $\dim H^1 = 0$ , so the relation hypergraph isolates non-trivial topology from the same activations.

**Heterogeneous restriction maps and holonomy.** Per-node  $P_v$  with Procrustes transports  $Q_{uv} \neq I$  break Proposition 3.1 (Remark 3.2). Cycle holonomy obstructions on the relation hypergraph are 2.5–3.1, below the Haar RMS value  $\sqrt{2k} = 4$ , and no architecture deviates significantly from a shuffled-paraphrase null (one-sided  $p = 0.10$ – $0.33$ , Appendix B.12). A node-space operationalization of  $H^1$  did not transfer to retrieval or steering (Appendix B.13). Both cycle-bearing constructions, depth-induced grid and relation-induced hypergraph, lift the framework off the matching-graph  $H^1 = 0$  floor.

## 5 Experiments

### 5.1 Setup and Datasets

We evaluate on nine transformer models spanning architecture families (GPT-2 [51], Llama [59, 26], Mistral [38], Qwen [65], phi [42], Gemma [24], OLMo [49]) and scales (124M to 13B parameters). Functional validation ablates coordinates on 1000 sentences (four models; 666 MRPC and 334 PAWS) or 100 MRPC sentences (five models). Restriction maps (PCA, CCA, contrastive Fisher) are learned from 200 pairs of the Microsoft Research Paraphrase Corpus (MRPC) [16]. Steering uses the CounterFact benchmark [47] ( $n = 1000$  per model), and fact retrieval uses ROME CounterFact: 500 facts split 70/30 for the held-out comparison (Section 5.2), with a 60-fact, 5-paraphrase subset for the topology constructions. We extract mean-pooled hidden states at layer  $\ell \approx 2L/3$ , set edge dimension  $k = 128$  (capturing  $\geq 98\%$  paired variance), take the bottom and top 20 eigenvectors as  $H^0$  and  $H^1$  directions, and norm-match all steering vectors to  $\alpha = 0.3\|\bar{h}\|$ . Bootstrap CIs use 1000 resamples, McNemar’s test [46] handles paired comparisons, and Cohen’s  $d$  [11] reports effect sizes.

### 5.2 Content from the Kernel: $H^0$ Beats LEACE, and Joint Paraphrase Constraints Improve It

Paraphrase-invariance does not by itself imply fact discriminability: an invariant subspace could merge Paris–France with Berlin–Germany. Hard same-relation retrieval tests exactly this residual,

Table 2: Held-out CounterFact retrieval (top-1 accuracy, hard same-relation distractors,  $k = 20$  dims, 70/30 train/test split,  $n_{\text{test}} = 150$  facts). Sheaf  $H^0$  basis and LEACE eraser are fit on training facts only, and evaluation is on held-out facts not seen during fitting. Across eight architectures sheaf  $H^0$  at 20 dimensions beats LEACE preserved at full hidden dimension, with gaps ranging from +1.5pp (OLMo-2) to +42.2pp (GPT-2).

| Model       | Full | $H^0$ (20D) | LEACE-full | LEACE-20D | PCA-20 | Rand-20 | $\Delta$ |
|-------------|------|-------------|------------|-----------|--------|---------|----------|
| GPT-2       | 43.1 | <b>85.3</b> | 43.1       | 28.9      | 24.6   | 39.1    | +42.2    |
| Mistral-7B  | 77.7 | <b>87.1</b> | 77.7       | 36.8      | 34.5   | 59.2    | +9.4     |
| OLMo-2-7B   | 81.3 | <b>82.8</b> | 81.3       | 40.9      | 35.7   | 62.3    | +1.5     |
| Phi-2       | 69.1 | <b>87.1</b> | 69.1       | 38.8      | 37.2   | 55.1    | +18.0    |
| Qwen-2.5-7B | 49.4 | <b>64.1</b> | 49.4       | 36.8      | 37.5   | 43.7    | +14.7    |
| Gemma-2-9B  | 54.4 | <b>77.9</b> | 54.4       | 32.3      | 27.5   | 43.5    | +23.5    |
| Llama-2-7B  | 68.2 | <b>80.4</b> | 68.2       | 25.7      | 22.3   | 50.7    | +12.2    |
| Llama-3-8B  | 72.7 | <b>88.0</b> | 72.7       | 28.2      | 24.4   | 55.9    | +15.3    |

against baselines that are themselves low-dimensional or invariance-targeted. We compare sheaf  $H^0$  (20D) to LEACE’s preserved subspace on ROME CounterFact, splitting 500 facts 70/30 into train and held-out test ( $n_{\text{test}}=150$ ). LEACE is fit with paraphrase-vs-non-paraphrase as the erased binary concept; its eraser  $P^*$  is an oblique rank- $(d-1)$  projection, and we call  $\text{im } P^*$  the LEACE-preserved subspace, at full hidden dimension, while sheaf  $H^0$  is 20-dimensional and built without binary labels. Easy retrieval is top-1 across all held-out centroids, and hard retrieval restricts to same-relation distractors. Classical subspace and metric-learning baselines under the identical protocol appear in Appendix B.5:  $H^0$  beats every label-free alternative and trails the fact-labelled LDA and NCA on five of eight models.

On hard same-relation retrieval (Table 2), sheaf  $H^0$  at 20 dims beats LEACE-preserved at full hidden dim on all eight architectures (mean gap +17.1pp). The +42.2pp GPT-2 gap matches the smallest model, since smaller residual streams concentrate paraphrase content into lower-dimensional shared subspaces. On 7–9B models,  $H^0$  at 20 dims still exceeds LEACE at 4096 dims by 1.5–23.5pp.

**Dimension-controlled LEACE.** LEACE for a binary concept removes a single direction (its null space has rank 1, so the eraser  $P^*$  has rank  $d-1$ ) [5], so the dim-controlled 20D baseline (LEACE-20D, Table 2) is the top-20 PCs of the LEACE-preserved activations. Sheaf  $H^0$  beats LEACE-20D by 27–60 pp on every architecture (mean +48.0 pp). LEACE-20D itself lands within 6 pp of PCA-20, confirming the algebraic prediction that a rank-1 erasure leaves the top-PC ranking essentially unchanged. At matched dimensionality the gap is therefore attributable to the decomposition.

**The kernel selection drives the result; the projection  $P$  is interchangeable.** Table 11 ablates  $H^0$  on the same activations under three  $P$  regimes: PCA- $P$  (current), random orthonormal  $P$  (5 seeds), and identity  $P$  (bottom-20 eigenvectors of  $C_W$  in full  $d$ -dim native space). Mean across eight architectures: PCA- $P$  81.6%, random- $P$  74.2%, identity- $P$  77.5% vs. PCA-20-without-kernel 30.5% and Random-20 51.2%. With identity  $P$  this is the bottom eigenspace of  $C_W$  itself, so the content lives in the kernel of the within-pair difference covariance whatever  $P$  is; PCA reduces the eigenproblem from  $\mathbb{R}^d$  to  $\mathbb{R}^{128}$  at a cost of at most 7.4 pp. Non-learned restriction maps likewise match learned ones on heterophilic-graph benchmarks [34]; closed-form learned restriction maps [15] are complementary.

**Joint paraphrase constraints improve  $H^0$ .** The result above uses a pair-only graph, in which each fact contributes a chain of paraphrase pairs. Replacing it with a ring over all of a fact’s paraphrases ( $b_1 = N$  instead of 0), with activations and  $P$  unchanged, asks for invariance across every paraphrase of a fact jointly. Across eight architectures ring- $H^0$  beats pair- $H^0$  by +3.2 to +9.5 pp (mean +6.5 pp, Table 10); a clique matches the ring within 0.1 pp. An edge-count-matched control attributes the gain to the joint constraint: an acyclic path over six paraphrases, with the ring’s five edges and  $b_1 = 0$ , recovers it (+7.7 pp over pairs), and the ring exceeds that path by +0.5 pp, significant on none of the eight architectures (Appendix B.3). The cycle-closing difference is the sum of the path differences, so it adds no constraint on exact global sections:  $H^0$  is set by which paraphrases are connected, and cycles enter through the harmonic subspace of Section 4.1.

Table 3: Functional validation: output change from ablating the  $H^1$  coordinates over that from ablating control coordinates of matched per-coordinate variability. <sup>a</sup> 1000 sentences (666 MRPC, 334 PAWS), 64 coordinates, one layer ( $L/2$ ), effect on final-position logits. <sup>b</sup> 100 sentences, 20 coordinates, three layers ( $L/4$ ,  $L/2$ ,  $3L/4$ ), effect on the mean-pooled final hidden state, best of the three depths shown. CIs are bootstrap 95%. Cohen’s  $d$  uses the pooled standard deviation of the two effect distributions.

| Model                     | Ratio        | Layer | 95% CI       | Cohen’s $d$ |
|---------------------------|--------------|-------|--------------|-------------|
| GPT-2 (124M) <sup>a</sup> | 18.2×        | L6    | [17.2, 19.3] | 2.34        |
| Phi-2 <sup>b</sup>        | 14.48×       | L8    | [14.0, 15.0] | 13.98       |
| Mistral-7B <sup>b</sup>   | 19.71×       | L8    | [19.1, 20.3] | 18.45       |
| Qwen-2.5-7B <sup>b</sup>  | 5.76×        | L7    | [5.38, 6.12] | 6.63        |
| OLMo-2-7B <sup>b</sup>    | 9.17×        | L16   | [8.85, 9.49] | 12.93       |
| Gemma-2-9B <sup>b</sup>   | 9.24×        | L21   | [8.94, 9.54] | 22.02       |
| Llama-2-7B <sup>a</sup>   | 18.2×        | L16   | [17.3, 19.1] | 4.44        |
| Llama-3-8B <sup>a</sup>   | 5.62×        | L16   | [5.52, 5.71] | 9.08        |
| Llama-2-13B <sup>a</sup>  | <b>26.5×</b> | L20   | [26.1, 26.8] | <b>14.3</b> |

### 5.3 Influence from the Spectral Complement: $H^1$ Carries Concentrated Causal Influence

We ablate the  $H^1$  directions through a coordinate-wise proxy and compare against control coordinates. At layer  $\ell$  we rank the  $d$  residual-stream coordinates by their mean absolute within-pair difference  $\mathbb{E}|h_{i,j}^{(1)} - h_{i,j}^{(2)}|$  over MRPC paraphrase pairs; this is the diagonal analogue of  $C_W$ , and so a coordinate-axis proxy for the top eigenspace of  $L_{\mathcal{F}}$  under the identity restriction map (Table 11 shows that map recovers most of the  $H^0$  result). The top  $m$  coordinates form  $H^1$ , the bottom  $m$  form  $H^0$ , and the control is the  $m$  coordinates, excluding both sets, whose individual variability is closest to the mean variability of the  $H^1$  coordinates. We zero the chosen coordinates in the layer- $\ell$  output, run the rest of the forward pass, and measure the  $L_2$  change in the model output: the final-position logits in the 1000-sentence runs ( $m = 64$ , one layer at  $L/2$ ) and the mean-pooled final hidden state in the 100-sentence runs ( $m = 20$ , layers  $L/4$ ,  $L/2$ ,  $3L/4$ ). The ratio is mean effect under  $H^1$  ablation over mean effect under control ablation, with a bootstrap CI. The control is matched coordinate by coordinate, and its total variability is only 0.13 to 0.53 of the  $H^1$  set’s. The ratio therefore compares coordinates of matched variability, and part of it reflects the larger total variability the  $H^1$  set removes [45, 64]. We use “causal” in the activation-patching sense [62, 23, 8]. A projection ablation of the sheaf  $H^1$  eigenbasis itself shifts the next-token distribution 17.8–398× more than random subspaces of equal dimension on all nine models (median over layers, Appendix B.6). It also changes the generated text: fact recall collapses under  $H^1$  ablation and stays at baseline under matched random ablation, and  $H^1$  is the least dormant subspace we measure, which argues against the dormant-pathway reading of the subspace-patching illusion [45] (Appendix B.7).

The  $H^1$  coordinates exert 5.6–26.5× greater influence than the control coordinates across all nine models, and every 95% CI excludes 1 (Table 3, Figure 2). GPT-2 (124M) and Mistral-7B reach near-identical ratios (18.2× and 19.71×) despite the 56× size gap. Measured at three depths on all nine models (protocol *b*), the ratio peaks at  $L/4$  for seven and at  $L/2$  for OLMo-2-7B and Gemma-2-9B (Appendix B.9).

### 5.4 Fragility from Harmonic Mass: Sheaf Steering as an Architecture Diagnostic

Harmonic mass is computed from MRPC layer-transition differences and steering fragility from CounterFact fact preservation under random perturbation, so the two come from disjoint measurements of the same model. Fragility is the fraction of CounterFact facts preserved under a random direction norm-matched to 0.3 times the hidden-state norm at layer  $\lfloor 5L/8 \rfloor$  (Appendix A.3). The three lowest-mass architectures are the three most robust and the three highest are the three most fragile: Spearman  $\rho = 0.90$ , exact one-sided permutation  $p = 0.014$  over the 720 rankings. The one rank inversion, Qwen-2.5-7B against Llama-3-8B, is within sampling error (34.3% vs. 36.0%). Table 1 lists both quantities per model.

The rest of this subsection compares steering methods within a model. CounterFact evaluates whether steering preserves facts while changing style. We compute style direction as the difference in mean

Table 4: Steering on CounterFact ( $\alpha = 0.3$ ,  $n = 1000$  per model). Fact preservation (%) with 95% bootstrap CIs. McNemar  $p$  is paired against the best method in each block (bold). Style saturates at 100% for all methods on all models.

| Model       | Method                          | Fact%       | 95% CI       | McNemar $p$  |
|-------------|---------------------------------|-------------|--------------|--------------|
| Llama-2-7B  | Contrastive sheaf ( $H^1$ dims) | <b>31.0</b> | [28.2, 33.9] | –            |
|             | PIXEL [66]                      | 21.1        | [18.6, 23.9] | $< 10^{-16}$ |
|             | Full style (CAA)                | 13.7        | [11.7, 15.9] | $< 10^{-31}$ |
|             | $H^0$ -removed (sheaf)          | 13.7        | [11.6, 15.9] | $< 10^{-31}$ |
|             | $H^1$ dims (sheaf PCA)          | 12.1        | [10.0, 14.1] | $< 10^{-38}$ |
|             | Random                          | 4.2         | [3.0, 5.4]   | $< 10^{-70}$ |
| Llama-3-8B  | PIXEL [66]                      | <b>38.8</b> | [35.2, 42.4] | –            |
|             | Random                          | 36.0        | [33.0, 38.8] | 0.10         |
|             | Full style (CAA)                | 27.2        | [24.5, 29.9] | $< 10^{-15}$ |
|             | $H^1$ dims (sheaf PCA)          | 26.1        | [23.4, 28.7] | $< 10^{-2}$  |
|             | $H^0$ -removed (sheaf)          | 26.1        | [23.3, 28.8] | $< 10^{-2}$  |
| Mistral-7B  | CCA sheaf ( $H^1$ dims)         | <b>41.6</b> | [38.6, 44.4] | –            |
|             | Random                          | 39.4        | [36.3, 42.4] | 0.19         |
|             | PIXEL [66]                      | 34.8        | [31.1, 38.3] | $< 10^{-8}$  |
|             | Variance (top- $k$ )            | 33.0        | [30.0, 35.8] | $< 10^{-7}$  |
|             | Full style (CAA)                | 31.7        | [28.8, 34.6] | $< 10^{-9}$  |
|             | $H^1$ dims (sheaf PCA)          | 29.8        | [26.9, 32.6] | $< 10^{-12}$ |
| Llama-2-13B | $H^1$ dims (sheaf PCA)          | <b>32.7</b> | [29.8, 35.6] | –            |
|             | Random                          | 32.1        | [29.3, 35.0] | 0.74         |
|             | Variance (top- $k$ )            | 31.3        | [28.5, 34.1] | 0.82         |
|             | Full style (CAA)                | 29.6        | [26.8, 32.3] | 0.07         |

activations between target and original completions at layer  $\ell = \lfloor 5L/8 \rfloor$ , then add norm-matched perturbations. Style change is measured as the fraction of outputs differing from baseline, and fact preservation is the fraction containing the target entity (greedy decoding, 50 tokens). At  $\alpha = 0.3$ , style saturates at 100% for all methods on all models, so fact preservation is the key differentiator.

Table 4 shows an architecture-dependent pattern. On fragile Llama-2-7B, random steering collapses generation (4.2%), while directed methods preserve facts at 12–31%, led by the architecture-matched contrastive sheaf at 31.0% (McNemar  $p < 10^{-70}$  vs. random). On robust Mistral-7B the architecture-matched CCA sheaf numerically leads (41.6% vs. random 39.4%, McNemar  $p = 0.19$ , a statistical tie), and on Llama-3-8B random and PIXEL lead (36.0–38.8%); random competes at high  $\alpha$  on robust architectures, which is consistent with the concentrated influence of the  $H^1$  coordinates (Section 5.3) disrupting more at large magnitudes. Llama-2-13B occupies an intermediate regime where all methods cluster tightly (29.6–32.7%, McNemar  $p > 0.07$ ). Consistent with that reading, sheaf directions beat random at smaller magnitudes (below).

**Operating regime.**  $\alpha = 0.30$  is presented as a stress test. Sweeping  $\alpha \in [0.05, 0.50]$  (Appendix B.2),  $H^0$ -removed sheaf steering with the architecture-matched map beats random on both swept architectures (Mistral-7B and Llama-3-8B) at  $\alpha \in [0.10, 0.15]$  (mean +1.0–2.0 pp, with  $H^0$ -removed reaching 43% vs. random 41% on Mistral-7B at  $\alpha = 0.10$ ), and random becomes competitive from  $\alpha = 0.30$ . Use  $\alpha \in [0.10, 0.15]$  on robust architectures and  $\alpha \approx 0.30$  on fragile ones.

**Head-to-head with PIXEL.** PIXEL [66], which extracts steering directions from two views of the prompt and calibrates the magnitude per position, reproduces the architecture-dependent pattern (Table 4): 21.1% on Llama-2-7B (vs. 4.2% random) but 34.8% on Mistral-7B where random reaches 39.4%. PIXEL beats CAA on Llama-2-7B ( $p < 10^{-7}$ ), and contrastive sheaf leads both at 31.0%. The fragility pattern holds for both direction recipes, and our simplifications of PIXEL appear in Appendix B.11.

## 6 Related Work

This paper extends a non-archival workshop version [25] that contained only the single-layer construction with three pilot experiments; the multi-layer grids, relation hypergraphs ( $b_1 = 66$ ), harmonic-mass fragility diagnostic, held-out LEACE protocol, restriction-map ablation, PIXEL comparison, baseline suite, and self-supervised learning (SSL) connection are new.

**Sheaves in machine learning.** Hansen and Ghrist [31] developed the spectral theory of the sheaf Laplacian, which specializes to the connection Laplacian of Singer and Wu [56] on  $O(d)$ -bundles, and Hansen and Gebhart [29] introduced sheaf neural networks, since applied to heterophilic graphs [6, 2, 34], predictive coding [55], and restriction-map learning [15]. Duta et al. [17] build sheaf hypergraph networks, and Mule et al. [48] unify directed/undirected hypergraph Laplacians. In the closest concurrent work on transformers, Krasnovsky [41] proposes a sheaf-inconsistency score for circuit coherence and Javidnia [37] builds context-chart atlases with Fisher metrics and holonomy. We differ in stalks (paraphrase-graph nodes carrying hidden states under PCA/CCA/contrastive restrictions) and in target (post-hoc  $H^0/H^1$  decomposition for retrieval and steering).

**Subspace interpretability.** Sparse autoencoders [7, 12, 58, 20], probes [4], causal abstraction [22], CKA/SVCCA [40, 52], and distributed alignment search (DAS) and LEACE [23, 5] target individual features or whole-space comparison. The log det Cov regulariser of Section 7 shares its geometry with the decorrelation terms of VICReg [3] and Barlow Twins [67]. Invariance objectives (contrastive: SimCLR [9], InfoNCE [61]; non-contrastive: BYOL [27], Barlow Twins [67]; DCCA [1]) are training-time analogues that achieve paired-view agreement during training, while we read invariance off a frozen model via  $\ker(L_{\mathcal{F}})$ .

**Steering and linear representations.** Activation addition [60], representation engineering [69], and their variants [66, 63, 39] inject directions and are compared in a unified evaluation by Im and Li [35], while Da Silva et al. [14] document model-dependent steering reliability across 36 models and Tan et al. [57] document input-level unreliability, which Section 5.4 relates to harmonic mass. The linear-representation hypothesis [50, 28, 33] is qualified by Engels et al. [19] and Makelov et al. [45]; our coordinate ablations use per-coordinate variability-matched controls, which address that concern only in part (Section 5.3). We compare empirically to PIXEL (Section 5.4, Appendix B.11); Angular Steering [63] and MSRS [35] are application-side and orthogonal.

## 7 Discussion

Proposition 3.1 reduces  $L_{\mathcal{F}}$  to projected covariance for a single shared  $P$  with identity transports, here on the matching graph. Within that regime the restriction-map ablation locates content in the kernel of the within-pair difference covariance (Table 11), and joint multi-paraphrase constraints add +6.5 pp (Appendix B.3). Beyond it, multi-layer grids carry a model-derived harmonic mass whose ranking tracks steering fragility (Section 5.4). The mass needs only 30 MRPC pairs and one forward pass per sentence, so it can screen a model before a steering sweep. Graphs that join activations of different dimension (text and image encoders, or models of different width) need per-node maps  $P_v$  of different widths, which we leave to future work.

**Self-supervised connection.** A sheaf-regularized objective  $\mathcal{L} = \text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}(f)$  pairs VICReg’s invariance term ( $\text{tr } L_{\mathcal{F}}$  is the within-pair MSE up to scale) with a log-determinant spread term that, like VICReg’s variance and covariance terms [3] and Barlow Twins’ redundancy reduction [67], penalizes dimensional collapse, though not in VICReg’s hinge-plus-off-diagonal form; such regularizers are used alongside non-contrastive objectives such as BYOL and SimSiam [27, 10]. Across seven architectures the regulariser holds encoder rank at 57–60/64 where naive MSE collapses to 14–42/64 (per-architecture breakdown in Appendix B.14).

**Limitations.**  $P$  is architecture- and partially domain-dependent (Appendix B.1); operating regime is  $\alpha \in [0.10, 0.15]$ ; node-space  $H^1$  does not transfer and cycle holonomy does not differ significantly from a shuffled null (Appendices B.13, B.12); the fragility diagnostic is a rank correlation over six architectures ( $\rho = 0.90$ ), and more architectures are needed before it serves as a predictor. The construction needs only a grouping of contexts asserted to be equivalent, and a cross-lingual probe (Appendix B.8) shows it transfers to translation pairs; logical equivalence and dialogue continuation fit the same template but are untested.

## 8 Conclusion

We framed cross-context consistency as a sheaf condition: paraphrase pairs are local witnesses of agreement, the kernel of the sheaf Laplacian glues them into a content subspace, and its spectral complement holds the phrasing-dependent variation, all read off a frozen model without training. The kernel beats the preserved subspace of LEACE on held-out retrieval, the spectral complement concentrates causal influence, and the cycles the same operator exposes across layers give a harmonic mass whose ranking tracks steering fragility across six architectures ( $\rho = 0.90$ ).

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## References

- [1] Galen Andrew, Raman Arora, Jeff Bilmes, and Karen Livescu. Deep canonical correlation analysis. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 1247–1255, 2013.
- [2] Federico Barbero, Cristian Bodnar, Haitz Sáez de Ocáriz Borde, Michael Bronstein, Petar Veličković, and Pietro Liò. Sheaf neural networks with connection laplacians, 2022. URL <https://arxiv.org/abs/2206.08702>.
- [3] Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.
- [4] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- [5] Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [6] Cristian Bodnar, Francesco Di Giovanni, Benjamin Paul Chamberlain, Pietro Liò, and Michael M Bronstein. Neural sheaf diffusion: A topological perspective on heterophily and oversmoothing in GNNs. In *Advances in Neural Information Processing Systems*, volume 35, pages 18527–18541, 2022.
- [7] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.
- [8] Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Buck Shlegeris, and Nate Thomas. Causal scrubbing: A method for rigorously testing interpretability hypotheses. Alignment Forum / Redwood Research technical report, 2022. URL <https://www.alignmentforum.org/posts/JvZhhzyCHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing>.
- [9] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 1597–1607, 2020.
- [10] Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15750–15758, 2021.
- [11] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition, 1988.
- [12] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations*, 2024.

- [13] Justin Curry. Sheaves, cosheaves and applications. *arXiv preprint arXiv:1303.3255*, 2014.
- [14] Patrick Queiroz Da Silva, Hari Sethuraman, Dheeraj Rajagopal, Hannaneh Hajishirzi, and Sachin Kumar. Steering off course: Reliability challenges in steering language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 19856–19882, Vienna, Austria, 2025.
- [15] Leonardo Di Nino, Sergio Barbarossa, and Paolo Di Lorenzo. Learning sheaf laplacian optimizing restriction maps. In *2024 58th Asilomar Conference on Signals, Systems, and Computers*. IEEE, 2024. arXiv:2501.19207.
- [16] William B Dolan and Chris Brockett. Automatically constructing a corpus of sentential paraphrases. In *Proceedings of the Third International Workshop on Paraphrasing*, 2005.
- [17] Iulia Duta, Giulia Cassarà, Fabrizio Silvestri, and Pietro Lió. Sheaf hypergraph networks. In *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2309.17116>.
- [18] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.
- [19] Joshua Engels, Eric J Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations*, 2025.
- [20] Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025.
- [21] Thomas Gebhart, Jakob Hansen, and Paul Schrater. Knowledge sheaves: A sheaf-theoretic framework for knowledge graph embedding. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 206 of PMLR, pages 9094–9116, 2023.
- [22] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. *Advances in Neural Information Processing Systems*, 34:9574–9586, 2021.
- [23] Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Proceedings of the Third Conference on Causal Learning and Reasoning*, volume 236 of *Proceedings of Machine Learning Research*, pages 160–187. PMLR, 2024.
- [24] Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.
- [25] Bryce Grant and Peng Wang. Gluing local contexts into global meaning: A sheaf-theoretic decomposition of transformer representations. In *ICLR 2026 Workshop on Unifying Concept Representation Learning*, 2026. URL <https://openreview.net/forum?id=eub5YrhExo>. Non-archival.
- [26] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [27] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [28] Wes Gurnee and Max Tegmark. Language models represent space and time. In *International Conference on Learning Representations*, 2024.
- [29] Jakob Hansen and Thomas Gebhart. Sheaf neural networks. In *NeurIPS 2020 Workshop on Topological Data Analysis and Beyond*, 2020.
- [30] Jakob Hansen and Robert Ghrist. Learning sheaf Laplacians from smooth signals. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5446–5450, 2019. doi: 10.1109/ICASSP.2019.8683709.

- [31] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3(4):315–358, 2019.
- [32] Xiaofei He and Partha Niyogi. Locality preserving projections. In *Advances in Neural Information Processing Systems*, volume 16, 2003.
- [33] Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. Linearity of relation decoding in transformer language models. In *International Conference on Learning Representations*, 2024. arXiv:2308.09124.
- [34] Ferran Hernandez Caralt, Guillermo Bernárdez Gil, Iulia Duta, Pietro Liò, and Eduard Alarcón Cot. Joint diffusion processes as an inductive bias in sheaf neural networks. In *Proceedings of the Geometry-grounded Representation Learning and Generative Modeling Workshop (GRaM)*, volume 251 of *Proceedings of Machine Learning Research*, pages 249–263, 2024.
- [35] Shawn Im and Yixuan Li. A unified understanding and evaluation of steering methods. *arXiv preprint arXiv:2502.02716*, 2025.
- [36] Shankar Iyer, Nikhil Dandekar, and Kornel Csernai. Quora question pairs. <https://www.kaggle.com/c/quora-question-pairs>, 2017.
- [37] Hossein Javidnia. A gauge theory of superposition: Toward a sheaf-theoretic atlas of neural representations. *arXiv preprint arXiv:2603.00824*, 2026.
- [38] Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7B. *arXiv preprint arXiv:2310.06825*, 2023.
- [39] Xinyan Jiang, Lin Zhang, Jiayi Zhang, Qingsong Yang, Guimin Hu, Di Wang, and Lijie Hu. Adaptive multi-subspace representation steering for attribute alignment in large language models. *arXiv preprint arXiv:2508.10599*, 2025.
- [40] Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529. PMLR, 2019.
- [41] Anatoly A. Krasnovsky. Measuring uncertainty in transformer circuits with effective information consistency. *Automatic Documentation and Mathematical Linguistics*, 59(Suppl 5):S423–S429, 2025.
- [42] Yuanzhi Li, Sébastien Bubeck, Ronen Eldan, Allie Del Giorno, Suriya Gunasekar, and Yin Tat Lee. Textbooks are all you need II: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023.
- [43] Lek-Heng Lim. Hodge laplacians on graphs. *SIAM Review*, 62(3):685–715, 2020. doi: 10.1137/18M1223101.
- [44] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3214–3252, Dublin, Ireland, 2022. Association for Computational Linguistics.
- [45] Aleksandar Makelov, Georg Lange, Atticus Geiger, and Neel Nanda. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *International Conference on Learning Representations*, 2024.
- [46] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [47] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*, volume 35, pages 17359–17372, 2022.
- [48] Emanuele Mule, Stefano Fiorini, Antonio Purificato, Federico Siciliano, Stefano Coniglio, and Fabrizio Silvestri. Directional sheaf hypergraph networks: Unifying learning on directed and undirected hypergraphs. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://arxiv.org/abs/2510.04727>.
- [49] OLMo Team. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025. COLM 2025.

- [50] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning*. PMLR, 2024.
- [51] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [52] Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [53] Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.828.
- [54] Michael T. Schaub, Yu Zhu, Jean-Baptiste Seby, T. Mitchell Roddenberry, and Santiago Segarra. Signal processing on higher-order networks: Livin’ on the edge... and beyond. *Signal Processing*, 187:108149, 2021.
- [55] Jeffrey Seely. Sheaf cohomology of linear predictive coding networks. In *NeurIPS 2025 Workshop on Symmetry and Geometry in Neural Representations*, 2025. arXiv:2511.11092.
- [56] Amit Singer and Hau-tieng Wu. Vector diffusion maps and the connection laplacian. *Communications on Pure and Applied Mathematics*, 65(8):1067–1144, 2012. doi: 10.1002/cpa.21395.
- [57] Daniel Tan, David Chanin, Aengus Lynch, Brooks Paige, Dimitrios Kanoulas, Adrià Garriga-Alonso, and Robert Kirk. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- [58] Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, et al. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024.
- [59] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- [60] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.
- [61] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- [62] Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL <https://arxiv.org/abs/2004.12265>.
- [63] Hieu M. Vu and Tan M. Nguyen. Angular steering: Behavior control via rotation in activation space. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2510.26243.
- [64] Zhengxuan Wu, Atticus Geiger, Jing Huang, Aryaman Arora, Thomas Icard, Christopher Potts, and Noah D. Goodman. A reply to makelov et al. (2023)’s “interpretability illusion” arguments. *arXiv preprint arXiv:2401.12631*, 2024.
- [65] An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.
- [66] Manjiang Yu, Hongji Li, Priyanka Singh, Xue Li, Di Wang, and Lijie Hu. PIXEL: Adaptive steering via position-wise injection with eXact estimated levels under subspace calibration. *arXiv preprint arXiv:2510.10205*, 2025.
- [67] Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 12310–12320, 2021.

- [68] Yuan Zhang, Jason Baldridge, and Luheng He. PAWS: Paraphrase adversaries from word scrambling. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1298–1308, 2019.
- [69] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

## Appendix Contents

|                                                    |    |
|----------------------------------------------------|----|
| <b>A Theoretical Foundations</b>                   | 16 |
| A.1 Algebraic $H^1 \neq 0$ on Richer Topologies    | 16 |
| A.2 Simplicial Complex Variants and Scale-Up       | 17 |
| A.3 Harmonic Mass vs Steering Fragility            | 19 |
| A.4 Geometric Picture: Where $H^1$ Lives           | 19 |
| A.5 Proof of the Reduction and Spectral Validation | 19 |
| <b>B Additional Experiments</b>                    | 20 |
| B.1 Cross-Dataset Generalization                   | 20 |
| B.2 Steering Magnitude Sweep                       | 20 |
| B.3 Cycle-Aware $H^0$ (Multi-Paraphrase Rings)     | 20 |
| B.4 Restriction Map Sensitivity                    | 21 |
| B.5 Retrieval Baseline Comparison                  | 22 |
| B.6 Sheaf Eigenbasis Ablation                      | 23 |
| B.7 Dormancy and the Subspace-Patching Illusion    | 23 |
| B.8 Cross-Lingual Invariance Probe                 | 24 |
| B.9 Per-Architecture Layer Profiles                | 24 |
| B.10 McNemar Tests with BH Correction              | 24 |
| B.11 PIXEL Head-to-Head on CounterFact             | 25 |
| B.12 Holonomy Null Calibration                     | 25 |
| B.13 Node-Space $H^1$ Operationalizations          | 26 |
| B.14 SSL Collapse Prevention (Per-Architecture)    | 26 |
| B.15 MLST on STS-B (Full Table)                    | 26 |
| <b>C Reproducibility</b>                           | 27 |
| C.1 Hyperparameter Sensitivity                     | 27 |
| C.2 Experimental Details                           | 28 |
| C.3 Practitioner Guidance                          | 29 |
| C.4 NeurIPS Paper Checklist                        | 29 |

## A Theoretical Foundations

### A.1 Algebraic $H^1 \neq 0$ on Richer Topologies

On the matching graph used in the main paper,  $H^1 = 0$  algebraically because the graph is acyclic and  $P^\top$  is surjective. To produce a topology with  $H^1 \neq 0$ , we construct three complexes from the same 300 activation vectors (60 facts, 5 paraphrases each):

1. Matching graph (paper baseline): 300 nodes, 120 edges (one per paraphrase pair), 0 faces. Algebraic  $H^1 = 0$ .
2. Fact clique (filled): 300 nodes, 600 edges (all pairs), 600 filled triangles. Each clique is simply connected, so  $H^1 = 0$  (though  $H^2 = 240$ ).
3. Relation hypergraph (unfilled cycles): 300 nodes, 360 edges (a five-node ring per fact, 300 edges, plus a ring per relation through the first paraphrase of each of its ten facts, 60 edges), 0 filled faces. Creates non-contractible cycles, yielding  $\dim H^1 = 66$ .

The relation rings join facts that share a relation (e.g., all country-capital facts). This complex carries cycles that no 2-simplex fills. Cohomology dimensions here are computed with trivial  $\mathbb{R}$  coefficients on the underlying simplicial complex; with learned Procrustes transports the sheaf cohomology of the induced learned- $Q$  sheaf can differ, and the cycle holonomies in Appendix B.12 are a separate probe of that twist.

**Hodge harmonic fraction.** On random unit edge cochains (uniform random direction per edge), we compute two fractions of cochain energy: the part in  $\ker((\delta^0)^\top)$ , the orthogonal complement of  $\text{im}(\delta^0)$ , and the harmonic part in  $\ker((\delta^0)^\top) \cap \ker(\delta^1) \cong H^1$ . The two coincide on face-free complexes ( $\delta^1 = 0$ ). For a random cochain the expected harmonic fraction is exactly  $b_1/|E|$ .

The relation hypergraph has expected harmonic fraction exactly  $b_1/|E| = 66/360 = 18.33\%$ , independent of architecture; the measured 18.4% is within probe noise. The filled cliques put 60% of

Table 5: Genuine cohomology and Hodge harmonic fraction across architectures and topologies.

| Topology            | $H^0$ | $H^1$ (algebraic) | $H^2$ | $\ker(\delta^0)^\top$ frac. | Harmonic frac. |
|---------------------|-------|-------------------|-------|-----------------------------|----------------|
| Matching (acyclic)  | 180   | 0                 | 0     | 0.000                       | 0.000          |
| Clique (filled)     | 60    | 0                 | 240   | 0.600                       | 0.000          |
| Relation hypergraph | 6     | <b>66</b>         | 0     | 0.184                       | <b>0.184</b>   |

the energy outside  $\text{im}(\delta^0)$ , but all of it is coexact (every cycle is a sum of triangle boundaries), so their harmonic fraction is zero.

#### Six-model LEACE table (whole 60-fact set).

Table 6: LEACE vs sheaf  $H^0$  on 60-fact retrieval (20 dimensions, top-1 against all 60 facts). LEACE is fitted on paraphrase versus non-paraphrase fact prompts, and sheaf  $H^0$  uses the joint-PCA restriction map.

| Model       | Full  | $H^0$ (20D)  | LEACE preserved | LEACE erased | PCA-20 | Random-20 |
|-------------|-------|--------------|-----------------|--------------|--------|-----------|
| GPT-2       | 85.3% | <b>95.7%</b> | 85.0%           | 2.0%         | 22.7%  | 53.7%     |
| Phi-2       | 97.3% | 93.0%        | 97.3%           | 3.3%         | 38.0%  | 64.3%     |
| Qwen-2.5-7B | 73.3% | <b>82.0%</b> | 75.0%           | 1.0%         | 30.3%  | 50.0%     |
| OLMo-2-7B   | 100%  | 97.0%        | 100%            | 2.3%         | 36.7%  | 77.3%     |
| Mistral-7B  | 100%  | 80.7%        | 100%            | 2.0%         | 50.0%  | 71.3%     |
| Gemma-2-9B  | 99.3% | 83.7%        | 99.3%           | 1.3%         | 26.7%  | 68.3%     |

On the same 300 vectors, Table 6 compares sheaf  $H^0$  with LEACE. Sheaf  $H^0$  outperforms LEACE preserved by up to 10.7pp on GPT-2 and Qwen-2.5-7B and outperforms full-space on those models despite using 20 of 768–3584 dimensions. LEACE makes the smallest least-squares edit that makes a binary concept (paraphrase vs non-paraphrase) linearly unpredictable; the sheaf identifies directions of minimal within-pair variation. LEACE-erased correctly destroys information (1–3%), validating the comparison.

## A.2 Simplicial Complex Variants and Scale-Up

The body uses 1-skeletons and edge dimension  $k = 8$ . We test whether its numbers depend on either choice: we fill the multi-layer grid into a 2-complex, fill the paraphrase cliques of the relation hypergraph, and rerun Table 1 at  $k = 64$  and  $k = 128$ .

**Triangulated multi-layer grid.** Section 4.1 explains why filling each square with two 2-simplices sends  $b_1$  and the harmonic mass to zero; this appendix reports the numbers on that 2-complex.

A diagonal edge from  $(a, l)$  to  $(b, l+1)$  carries  $\omega(e) = (P_{l+1}^\top h_b^{l+1} - \mu_{l+1}) - R_{l \rightarrow l+1}^\top (P_l^\top h_a^l - \mu_l)$ , the vertical-edge formula across the pair. The architecture-dependent quantity that survives is the *coexact fraction*: the energy of the orthogonal projection of  $\omega$  onto  $\text{im}(\delta^1)^\top$ , which is determined by the curl  $\delta^1 \omega$ . This is the model-implied 2-cochain on the filled triangles, equivalently the curl (circulation) of  $\omega$  around each filled triangle, i.e. the mismatch between the diagonal-then-horizontal and vertical-then-diagonal path sums.

The coexact fraction in Table 7 reproduces the split of Table 1 at every  $L \leq 8$ : the three robust architectures stay below 3% and the three fragile ones lie between 5% and 20%. The same content that the 1-skeleton’s harmonic mass picked up has migrated, in the 2-complex, into the coexact direction: the layer transitions still fail to commute with paraphrase equivalence on the same architectures, but now that failure decomposes into the part the 2-skeleton can absorb (coexact) rather than into a topological harmonic class.

**Triangulated relation hypergraph.** Filling each fact’s paraphrase 5-clique with all  $\binom{5}{3} = 10$  triangles makes each fact simply connected, so the 60 within-fact 1-cycles become 1-boundaries. Across the whole complex this gives  $b_0 = 6$  (one component per relation),  $b_1 = 6$  (the relation cycles, which are the only cycles that span more than one fact), and  $b_2 = 240$  (each  $K_5$  2-skeleton has  $b_2 = 4$ , and there are 60 facts). Random unit edge cochains have a measured Hodge harmonic

Table 7: Triangulated multi-layer grid (each square filled with two triangles sharing a diagonal). Edge dim  $k = 64$ , 30 MRPC paraphrase pairs.  $b_1 = b_2 = 0$  for every architecture and every  $L$ , so  $\omega$  has no harmonic component (Hodge harmonic mass collapses to machine precision). Reported quantities: model-derived  $\omega$  norm squared, coexact fraction (energy explained by the 2-skeleton), and face-cochain norm squared  $\|\delta^1 \omega\|^2$ . Mistral-7B and Llama-3-8B place under 1% on the 2-skeleton and Qwen-2.5-7B under 3%, while GPT-2, Phi-2 and Llama-2-7B place 5–20%.

| Architecture | $L = 2$ |                         | $L = 4$ |                         | $L = 8$ |                         |
|--------------|---------|-------------------------|---------|-------------------------|---------|-------------------------|
|              | coexact | $\ \delta^1 \omega\ ^2$ | coexact | $\ \delta^1 \omega\ ^2$ | coexact | $\ \delta^1 \omega\ ^2$ |
| Mistral-7B   | 0.0000  | $1.4 \times 10^2$       | 0.0002  | $8.6 \times 10^2$       | 0.0006  | $2.5 \times 10^3$       |
| Llama-3-8B   | 0.0005  | $4.7 \times 10^2$       | 0.0026  | $1.8 \times 10^3$       | 0.0075  | $5.1 \times 10^3$       |
| Qwen-2.5-7B  | 0.0003  | $9.8 \times 10^2$       | 0.0255  | $6.0 \times 10^5$       | 0.0247  | $9.7 \times 10^5$       |
| GPT-2 (124M) | 0.0544  | $6.7 \times 10^3$       | 0.1219  | $1.4 \times 10^5$       | 0.1014  | $2.2 \times 10^5$       |
| Phi-2        | 0.0572  | $5.1 \times 10^3$       | 0.1895  | $7.9 \times 10^4$       | 0.1970  | $2.0 \times 10^5$       |
| Llama-2-7B   | 0.0759  | $6.4 \times 10^4$       | 0.0815  | $8.6 \times 10^4$       | 0.0623  | $1.3 \times 10^5$       |

mass of 0.93% on this 2-complex (expected  $6/660 = 0.91\%$ ), down from 18.4% on the 1-skeleton ring (Table 5); the Hodge mass is determined by the topology and the  $k = 64$  stalk dimension and is identical across architectures by construction.

The 6 surviving harmonic 1-forms are exactly the relation cycles. The new  $b_2 = 240$  harmonic 2-forms do not pull back to interpretable structure on the residual stream: the model-implied 2-cochain  $\delta^1 \omega$  is zero to machine precision under shared-PCA restriction maps (every transport is identity, every cochain is exact via the node potential  $P^\top h_v$ ), so the 2-complex carries no degree-2 cohomological obstruction beyond what the 1-skeleton already encodes. With per-node restriction maps and learned Procrustes transports the situation can change; we leave the heterogeneous-restriction  $H^2$  probe to future work.

**Edge-dimension scale-up.** The body Table 1 uses  $k = 8$ . To check whether the variation across architectures, their ordering by Hodge mass, and the body numbers hold at higher  $k$ , we rerun the same multi-layer grid construction at  $k = 64$  ( $N = 30$ ) and at  $k = 128$  ( $N = 80$ , since  $k$  is capped by  $\min(2N, k_{\text{requested}})$ ).

Table 8: Multi-layer matching grid Hodge harmonic mass across edge dimensions. The body Table 1 reports  $k = 8$ . Increasing  $k$  keeps the three robust architectures below the three fragile ones at every  $L \leq 8$  and widens the span to more than three orders of magnitude. The  $k = 128$  row uses  $N = 80$  paraphrase pairs to make the requested  $k$  admissible (otherwise  $k$  is capped at  $\min(2N, k_{\text{req}}) = 60$ ).

| Architecture | $k = 64$ ( $N = 30$ ) |         |         | $k = 128$ ( $N = 80$ ) |         |         |
|--------------|-----------------------|---------|---------|------------------------|---------|---------|
|              | $L = 2$               | $L = 4$ | $L = 8$ | $L = 2$                | $L = 4$ | $L = 8$ |
| Mistral-7B   | 0.00003               | 0.0002  | 0.0005  | 0.00002                | 0.0002  | 0.0005  |
| Llama-3-8B   | 0.0005                | 0.0022  | 0.0061  | 0.0004                 | 0.0017  | 0.0049  |
| Qwen-2.5-7B  | 0.0003                | 0.0234  | 0.0248  | 0.0003                 | 0.0191  | 0.0215  |
| GPT-2 (124M) | 0.050                 | 0.112   | 0.101   | 0.041                  | 0.081   | 0.084   |
| Phi-2        | 0.054                 | 0.180   | 0.197   | 0.053                  | 0.156   | 0.180   |
| Llama-2-7B   | 0.067                 | 0.077   | 0.065   | 0.083                  | 0.093   | 0.078   |

At every  $L \leq 8$  and every  $k$  the three robust architectures (Mistral-7B, Llama-3-8B, Qwen-2.5-7B) stay below the three fragile ones (GPT-2, Phi-2, Llama-2-7B), and Mistral-7B is the lowest (tied with Qwen-2.5-7B at  $k = 8$ ,  $L = 2$ ). The span widens to more than three orders of magnitude ( $\approx 2 \times 10^{-5}$  to  $8 \times 10^{-2}$ ). Within the fragile three the order shifts with  $k$  and  $L$ : Llama-2-7B is highest at  $L = 2$ , and Phi-2 is highest at  $L \geq 4$ . The body’s  $L = 2$  values ( $\approx 10^{-4}$  on Mistral-7B and Qwen-2.5-7B,  $\approx 10^{-3}$  on Llama-3-8B,  $\approx 0.08$  on Llama-2-7B) hold at every edge dimension.

### A.3 Harmonic Mass vs Steering Fragility (Per-Model)

Table 1 lists the two quantities behind the test in Section 5.4: multi-layer Hodge harmonic mass ( $L = 2$ ) and steering fragility. Fragility is fact preservation on CounterFact under a random direction norm-matched to 0.3 times the hidden-state norm at layer  $\lfloor 5L/8 \rfloor$  (lower means more fragile),  $n = 1000$  prompts per model (200 for Phi-2), scored with the same criterion as Table 4. The two quantities come from disjoint measurements (MRPC layer transitions vs. CounterFact steering). The rank correlation is  $\rho = 0.90$  (exact one-sided permutation  $p = 0.014$  over 720 rankings), and the sole rank inversion, Qwen-2.5-7B vs. Llama-3-8B, is within sampling error (34.3% vs. 36.0% at  $n = 1000$ ).

### A.4 Geometric Picture: Where $H^1$ Lives

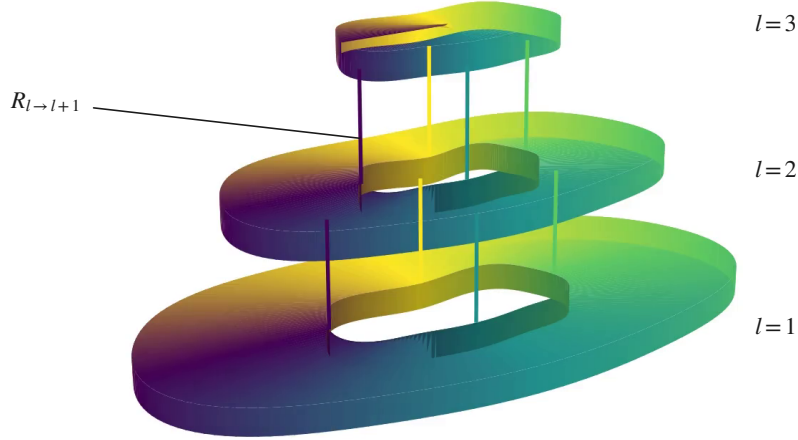


Figure 3: Schematic of the multi-layer matching sheaf, a frame from an animation. Each slab is one layer  $l$ , and struts join the same sentence at adjacent layers through the Procrustes transport  $R_{l \rightarrow l+1}$ . A path up one strut, across layer  $l + 1$ , and down another strut closes an inter-layer square. These squares give the grid its  $b_1 = N(L - 1)$  independent cycles and carry the harmonic mass. The slab shapes are illustrative and carry no topology of their own.

### A.5 Proof of the Reduction and Spectral Validation

**Proof of Proposition 3.1.** On the matching graph with a single shared  $P$  and identity transports, each edge cochain is  $\Delta_i^{(k)} = P^\top h_i^{(1)} - P^\top h_i^{(2)} = P^\top \Delta_i$  with  $\Delta_i = h_i^{(1)} - h_i^{(2)}$ . Hence

$$L_{\mathcal{F}} = \frac{1}{N} \sum_i \Delta_i^{(k)} (\Delta_i^{(k)})^\top = P^\top \left( \frac{1}{N} \sum_i \Delta_i \Delta_i^\top \right) P = P^\top C_W P,$$

which is (i). For unit  $v$ ,  $v^\top L_{\mathcal{F}} v = \frac{1}{N} \sum_i (v^\top P^\top \Delta_i)^2$ , which is (iii). The Fisher baseline replaces the standard eigenproblem with the generalized one  $L_{\mathcal{F}} v = \lambda (P^\top C_T P) v$ , which is (ii).  $\square$  For Remark 3.2: with per-node maps the edge cochain is  $c_e = P_v^\top h_v - Q_{uv} P_u^\top h_u$ , and  $\frac{1}{N} \sum_e c_e c_e^\top$  contains the cross-terms  $P_v^\top h_v h_u^\top P_u Q_{uv}^\top$ , which do not factor through a single  $P$ .

**Numerical sanity check.** In exact arithmetic the assembled Laplacian  $(\delta^0)^\top \delta^0$  on  $C^0$  has  $\dim \ker = \dim C^0 - \text{rank}(\delta^0)$ . We check that the eigenvalue-threshold estimate matches the rank-based one on GPT-2 with 50 MRPC pairs (edge dimension  $k = 49$ , the cap  $\min(64, d, n - 1)$  used by the script):

- Spectral  $H^0$  dimension (eigenvalues  $< 10^{-6}$ ): 49
- Algebraic  $H^0$  dimension ( $\dim C^0 - \text{rank}(\delta^0)$  on the assembled coboundary matrix): 50
- Coboundary test:  $\|\delta^0 \Pi_{H^0} s\| < 10^{-6}$  for all features tested

The two estimates agree up to one near-threshold eigenvalue ( $\lambda \approx 4 \times 10^{-6}$ , just above the  $10^{-6}$  cutoff).

## B Additional Experiments

### B.1 Cross-Dataset Generalization

We test out-of-domain  $P$  transfer for  $H^0$  retrieval on held-out CounterFact (Table 9): fit the joint-PCA  $P$  on MRPC, PAWS, or QQP paraphrase pairs ( $n = 300$ ) instead of CounterFact training facts, then apply it to identical CounterFact test centroids and queries. Across eight architectures, in-domain  $P$  averages 81.6%, out-of-domain  $P$  averages 55–60%, PCA-20 (no kernel) 30.5%, and Random-20 51.2%. Cross-domain  $P$  retains kernel-selection benefit over no-kernel baselines (+5 to +30 pp) but loses  $\sim 25$  pp relative to in-domain. The within-pair structure that  $P$  captures is therefore partially domain-specific: news paraphrases (MRPC), adversarial near-duplicates (PAWS), and question paraphrases (QQP) all transfer non-trivially, but a fact-distinguishing retrieval task benefits from fact-domain  $P$ . This is a limitation, and we recommend in-domain  $P$  when the downstream retrieval domain is fixed.

Table 9: Cross-dataset restriction-map transfer (held-out CounterFact, hard same-relation retrieval,  $k = 20$ ).  $H^0$  is fit with  $P$  from the source on the left and evaluated on the same CounterFact test set. Out-of-domain  $P$  retains +5 to +30 pp over no-kernel baselines but loses  $\sim 25$  pp relative to in-domain  $P$ .

| Model       | $H^0$ CF (in-domain) | $H^0$ MRPC | $H^0$ PAWS | $H^0$ QQP | PCA-20 | Random-20 |
|-------------|----------------------|------------|------------|-----------|--------|-----------|
| GPT-2       | 85.3                 | 55.1       | 63.9       | 59.1      | 24.6   | 39.1      |
| Phi-2       | 87.1                 | 57.3       | 61.6       | 57.6      | 37.2   | 55.1      |
| Mistral-7B  | 87.1                 | 57.8       | 65.9       | 61.2      | 34.5   | 59.2      |
| OLMo-2-7B   | 82.8                 | 64.1       | 60.5       | 57.6      | 35.7   | 62.3      |
| Qwen-2.5-7B | 64.1                 | 47.0       | 53.3       | 56.7      | 37.5   | 43.7      |
| Gemma-2-9B  | 77.9                 | 48.3       | 47.9       | 51.0      | 27.5   | 43.5      |
| Llama-2-7B  | 80.4                 | 60.5       | 60.0       | 53.5      | 22.3   | 50.7      |
| Llama-3-8B  | 88.0                 | 56.9       | 65.5       | 60.5      | 24.4   | 55.9      |
| <b>Mean</b> | <b>81.6</b>          | 55.9       | 59.8       | 57.1      | 30.5   | 51.2      |

### B.2 Steering Magnitude Sweep

We sweep  $\alpha \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$  on Mistral-7B and Llama-3-8B. At moderate magnitudes ( $\alpha=0.10$ – $0.15$ ), sheaf-derived methods outperform random steering on both models (+1.0–2.0pp). At high magnitudes ( $\alpha \geq 0.30$ ), concentrated  $H^1$  influence causes over-perturbation on robust models, and random becomes competitive. This tradeoff drives the practical recommendations in Appendix C.3.

### B.3 Cycle-Aware $H^0$ : Multi-Paraphrase Rings

The body retrieval pipeline uses a pair-only graph (each fact contributes a chain of  $K-1$  paraphrase edges; the resulting forest has  $b_1 = 0$ ). We test whether constraining all paraphrases of a fact jointly gives better  $H^0$  for retrieval, and whether any gain comes from the cycle itself.

**Construction.** For each fact with  $K \in [4, 7]$  paraphrases (capped at 5), we build a *multi-paraphrase ring*: a closed cycle on the  $K$  paraphrases. With  $N_{\text{train}} = 350$  facts, the ring complex has  $\sum_i K_i$  edges and  $b_1 = N_{\text{train}}$  cycles ( $\sim 340$  cycles). The kernel of  $L_{\mathcal{F}}$  on this ring requires invariance *across all paraphrases of each fact*, not just pairwise. Pair-only and ring share the same node activations and the same joint-PCA  $P$ ; only the edge set (and therefore the cycle structure) differs.

Both constructions win on every architecture, mean +6.5 pp over pair-only, and ring and clique differ by 0.1 pp on average, so edge density adds nothing beyond the ring.

**Edge-count-matched control.** To separate the joint constraint from the cycle, we compare the five-node ring (5 edges,  $b_1 = 1$ ) with an acyclic path over six paraphrases of the same fact (5 edges,  $b_1 = 0$ ), on the pool of facts with at least six paraphrases, with paired bootstrap CIs over held-out facts. The path improves over pairs by +7.7 pp on average across the eight architectures, and the ring

Table 10: Cycle-aware  $H^0$  for hard-CounterFact retrieval. Both multi-paraphrase rings ( $b_1 = N$ ) and full paraphrase cliques ( $b_1 = N(K-1)(K-2)/2$ ) beat the pair-only acyclic forest ( $b_1 = 0$ ) on all eight architectures, mean +6.5 pp. Ring and clique perform within 0.1 pp of each other.

| Model       | $H^0$ pair ( $b_1=0$ ) | $H^0$ ring ( $b_1=N$ ) | $H^0$ clique ( $b_1 \gg N$ ) | $\Delta_{\text{ring}}$ | $\Delta_{\text{clique}}$ |
|-------------|------------------------|------------------------|------------------------------|------------------------|--------------------------|
| GPT-2       | 87.4                   | 90.7                   | 89.2                         | +3.4                   | +1.8                     |
| Phi-2       | 83.1                   | 89.6                   | 90.5                         | +6.5                   | +7.4                     |
| Mistral-7B  | 86.7                   | 89.8                   | 91.4                         | +3.2                   | +4.7                     |
| OLMo-2-7B   | 84.4                   | 91.4                   | 91.4                         | +7.0                   | +7.0                     |
| Qwen-2.5-7B | 62.3                   | 71.8                   | 70.7                         | +9.5                   | +8.4                     |
| Gemma-2-9B  | 75.2                   | 84.0                   | 84.7                         | +8.8                   | +9.5                     |
| Llama-2-7B  | 80.8                   | 88.9                   | 89.6                         | +8.1                   | +8.8                     |
| Llama-3-8B  | 86.5                   | 92.3                   | 90.7                         | +5.9                   | +4.3                     |
| <b>Mean</b> | <b>80.8</b>            | <b>87.3</b>            | <b>87.3</b>                  | <b>+6.5</b>            | <b>+6.5</b>              |

exceeds the path by +0.5 pp on average, with the 95% CI including zero on all eight. The gain is therefore carried by constraining all of a fact’s paraphrases jointly; the cycle-closing difference is the sum of the path differences and adds no constraint on exact global sections.

#### B.4 Restriction Map Sensitivity

We evaluate three restriction maps: PCA (parameter-free), CCA (correlation-maximizing), and contrastive (Fisher discriminant). The optimal map is architecture-dependent: CCA gives +2.2pp on Mistral-7B ( $n=1000$ ,  $\alpha=0.3$ ), while contrastive gives 31.0% vs 4.2% on Llama-2-7B ( $7.4\times$  improvement,  $p < 10^{-70}$ ). Figure 4 compares three of the four CounterFact-evaluated architectures (Llama-2-13B’s smaller method grid appears only in Table 4) across Random, CAA, PIXEL, sheaf  $H^1$ , and the architecture-matched best sheaf method; the architecture-dependent pattern (sheaf wins outright on fragile, and is statistically tied with random on robust) is visible across rows.

**Restriction-map ablation: PCA vs random vs identity.** Table 11 reports the full restriction-map ablation summarised in Section 5.2. We compute  $H^0$  on the same activations under three regimes: PCA- $P$  (current method, joint-PCA top-128 PCs), random orthonormal  $P$  (5 seeds), and identity  $P$  (no projection; bottom-20 eigenvectors of  $C_W$  in the full  $d$ -dim residual stream). Hard-CounterFact retrieval, top-1 accuracy with same-relation distractors,  $n_{\text{test}} = 150$  facts.

Table 11: Restriction-map ablation (8 architectures, hard-CounterFact retrieval).  $H^0$  from  $\ker L_{\mathcal{F}}$  recovers content directions whether  $P$  is joint-PCA, a random orthonormal projection, or absent.

| Model       | $H^0$ PCA- $P$ | $H^0$ random- $P$ | $H^0$ identity- $P$ | PCA-20 (no kernel) | Random-20 |
|-------------|----------------|-------------------|---------------------|--------------------|-----------|
| GPT-2       | 85.3           | $77.1 \pm 0.8$    | 79.0                | 24.6               | 39.1      |
| Phi-2       | 87.1           | $73.0 \pm 2.0$    | 78.8                | 37.2               | 55.1      |
| Mistral-7B  | 87.1           | $80.6 \pm 1.6$    | 83.3                | 34.5               | 59.2      |
| OLMo-2-7B   | 82.8           | $76.4 \pm 1.6$    | 79.7                | 35.7               | 62.3      |
| Qwen-2.5-7B | 64.1           | $61.4 \pm 1.3$    | 62.3                | 37.5               | 43.7      |
| Gemma-2-9B  | 77.9           | $69.2 \pm 1.5$    | 76.5                | 27.5               | 43.5      |
| Llama-2-7B  | 80.4           | $76.3 \pm 2.7$    | 80.1                | 22.3               | 50.7      |
| Llama-3-8B  | 88.0           | $80.0 \pm 1.4$    | 79.9                | 24.4               | 55.9      |
| <b>Mean</b> | <b>81.6</b>    | <b>74.2</b>       | <b>77.5</b>         | 30.5               | 51.2      |

Identity- $P$  (no data-dependent projection) lands within 4.1 pp of PCA- $P$  on average, and beats PCA-without-kernel by +47 pp. Random- $P$  averaged over 5 seeds lands within 8 pp of PCA- $P$ . The kernel selection recovers the content, and the joint-PCA  $P$  shrinks the eigenproblem from  $\mathbb{R}^d$  to  $\mathbb{R}^{128}$ . On its own,  $P$  does not select content directions.

## B.5 Retrieval Baseline Comparison

We compare sheaf  $H^0$  to classical subspace and representation-learning baselines on the same task (Table 12): LDA, CCA, PLS, top-20 PCA, and metric learning (NCA), under the same held-out hard-CounterFact protocol as Table 2 (same activations, 70/30 fact split, 20-dimensional budget, top-1 scorer). This is an independent re-run: the sheaf  $H^0$  and Full columns differ from Table 2 by under 1 pp (seed and sampling), so every method here is scored on one consistent run. Whitened PCA (W-PCA) scales each of the top-20 principal directions by  $1/\sqrt{\lambda}$  on centred activations. Its column comes from a separate run under the same protocol, and whitening lifts PCA-20 by up to 17 pp but stays 27 to 48 pp below sheaf  $H^0$ . LDA and NCA use fact-identity labels; sheaf  $H^0$ , CCA, PLS, PCA-20, and W-PCA are label-free. Sheaf  $H^0$  beats every label-free alternative by +9 to +36 pp on all eight architectures and, on average, lands within  $\sim 2$  pp of the label-requiring discriminative ceiling (per-architecture gap up to 8 pp on Qwen), winning outright on Phi-2, Gemma-2, and Llama-3. Supervised sentence encoders (SimCSE 97.5%, a sentence-transformer 99.1%) import large-scale paraphrase supervision, so we list them only for reference. Unlike LDA and NCA, the sheaf operator also yields the  $H^1$  influence subspace, the steering decomposition, and the harmonic-mass fragility diagnostic from the same construction.

Table 12: Retrieval baseline comparison (held-out hard-CounterFact, top-1 %,  $k = 20$ , same protocol as Table 2). Label-free: sheaf  $H^0$ , CCA, PLS, PCA-20 (top-20 principal components without kernel selection, as in Table 2), W-PCA (the same 20 components whitened, each scaled by  $1/\sqrt{\lambda}$  on centred activations). Label-requiring: LDA, NCA. Sheaf  $H^0$  bold where it beats both LDA and NCA.

| Model       | Sheaf $H^0$ | LDA  | NCA  | CCA  | PLS  | PCA-20 | W-PCA | Full |
|-------------|-------------|------|------|------|------|--------|-------|------|
| GPT-2       | 84.9        | 87.4 | 88.3 | 47.9 | 49.0 | 24.4   | 39.5  | 43.1 |
| Phi-2       | <b>87.6</b> | 84.9 | 83.3 | 58.9 | 64.1 | 37.9   | 42.7  | 69.1 |
| Mistral-7B  | 86.9        | 90.1 | 88.3 | 60.7 | 63.7 | 34.3   | 45.1  | 77.4 |
| OLMo-2-7B   | 82.6        | 88.5 | 86.9 | 62.8 | 67.5 | 36.6   | 37.2  | 81.3 |
| Qwen-2.5-7B | 64.8        | 72.5 | 72.9 | 50.8 | 56.0 | 36.6   | 38.1  | 50.3 |
| Gemma-2-9B  | <b>77.4</b> | 76.1 | 76.3 | 51.2 | 50.1 | 27.3   | 37.2  | 54.4 |
| Llama-2-7B  | 80.6        | 83.5 | 84.4 | 49.7 | 49.9 | 21.9   | 39.3  | 68.2 |
| Llama-3-8B  | <b>87.8</b> | 86.5 | 86.0 | 53.3 | 55.5 | 24.4   | 39.3  | 72.7 |
| Mean        | 81.6        | 83.7 | 83.3 | 54.4 | 57.0 | 30.4   | 39.8  | 64.6 |

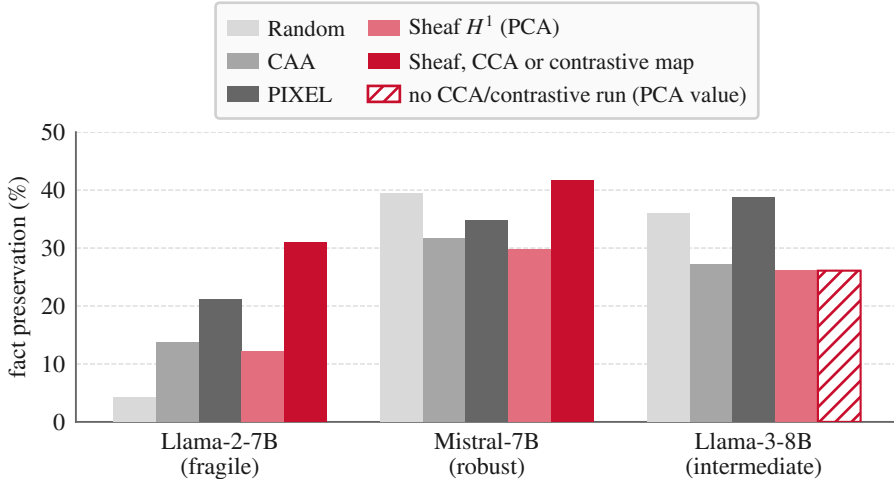


Figure 4: Steering on CounterFact at  $\alpha = 0.3$  ( $n = 1000$  per model) for the three architectures with a PIXEL run. The hatched bar on Llama-3-8B repeats the PCA  $H^1$  value, since that model has no CCA or contrastive run. On fragile Llama-2-7B, random steering collapses to 4.2% and the contrastive sheaf leads at 31.0%. On robust Mistral-7B, the CCA sheaf and random tie (41.6% vs. 39.4%). On Llama-3-8B, PIXEL and random lead. Same data as Table 4 and Appendix B.11.

## B.6 Sheaf Eigenbasis Ablation

This ablation uses the eigenbasis of  $L_{\mathcal{F}}$  directly. At every other layer,  $H^1$  is the top 20 eigenvectors of  $L_{\mathcal{F}}$  with the joint-PCA restriction map ( $k = 128$ ), fit on mean-pooled block outputs of 200 MRPC paraphrase pairs, and  $H^0$  is the bottom 20. On 100 held-out MRPC sentences, the block output is projected onto the orthogonal complement of the subspace at every position, the rest of the forward pass runs, and the effect is the KL divergence of the next-token distribution at the final position. The control is 20 Haar-random 20-dimensional subspaces, which are not matched to  $H^1$  in activation energy; Table 3 is the variability-matched comparison. Table 13 reports, per model, the ratio of mean KL ( $H^1$  over random) as the median and range over layers, the smallest fraction of prompts over layers for which  $H^1$  exceeds all 20 random draws, and the median  $H^0$  ratio.

Table 13: Sheaf eigenbasis ablation: ratio of mean next-token KL from projecting out  $H^1$  (top 20 eigenvectors of  $L_{\mathcal{F}}$ ) to that from 20 random 20-dimensional subspaces at every other layer. Median and range are over layers. The prompts column is the smallest fraction over layers of held-out prompts on which  $H^1$  exceeds all 20 draws.

| Model        | Layers | $H^1/\text{random}$ , median | Range     | Prompts | $H^0/\text{random}$ , median |
|--------------|--------|------------------------------|-----------|---------|------------------------------|
| GPT-2 (124M) | 6      | 17.8×                        | 13.7–53.5 | 99%     | 0.19                         |
| Phi-2        | 16     | 84.1×                        | 18.4–122  | 97%     | 0.56                         |
| Mistral-7B   | 16     | 398×                         | 70.0–3075 | 100%    | 0.54                         |
| Qwen-2.5-7B  | 14     | 127×                         | 78.7–293  | 99%     | 0.45                         |
| OLMo-2-7B    | 16     | 182×                         | 112–1048  | 100%    | 2.21                         |
| Gemma-2-9B   | 21     | 240×                         | 75.8–1211 | 100%    | 0.74                         |
| Llama-2-7B   | 16     | 287×                         | 102–2283  | 100%    | 0.32                         |
| Llama-3-8B   | 16     | 176×                         | 75.7–409  | 100%    | 0.29                         |
| Llama-2-13B  | 20     | 327×                         | 191–645   | 100%    | 0.47                         |

## B.7 Dormancy and the Subspace-Patching Illusion

Makelov et al. [45] describe an illusion for *interchange* patching,  $a + (p_A - p_B)v$  with  $v = (v_{\text{disc}} + v_{\text{dormant}})/\sqrt{2}$ , where variation along a causally disconnected direction activates a dormant one. Section 5.3 instead zero-ablates the subspace, projecting activations onto its orthogonal complement. Two checks rule out a dormant-pathway reading. A dormant direction is one that is not differentially active on-distribution, so we measure the on-distribution projection spread of  $H^1$ ,  $H^0$ , and five variance-matched random subspaces of the same dimension, norm, and eigenspectrum, then ablate each and read fact recall off the greedy 50-token continuation of the bare CounterFact prompt, which does not contain the answer ( $n = 100$  for GPT-2 and 150 for the 7B models).  $H^1$  directions are the least dormant subspace we measure, and ablating them collapses recall while the matched random ablation leaves it at baseline (Table 14); GPT-2 recalls only 15% of these facts unaided, so its row is the weakest. The generated text itself changes, so the effect is behavioural and does not rest on the tail of the KL distribution. On GPT-2 the intervention KL ratio is 15.7×; on the 7B models it saturates numerically (median raw KL 7.1 vs. 0.003 nats on Mistral-7B), so the recall collapse is the evidence we rely on there.

Table 14: Dormancy and behavioural ablation. Dormancy ratio is the on-distribution projection spread of  $H^1$  divided by the mean spread of variance-matched random subspaces. Recall is the fraction of greedy continuations of the bare prompt that contain the target entity. Layer  $\ell \approx 2L/3$ ,  $k = 20$ , five random draws, seed 42.

| Model      | Dormancy ratio ( $H^1/\text{random}$ ) | Recall, no ablation | $H^1$ ablated | Random ablated |
|------------|----------------------------------------|---------------------|---------------|----------------|
| GPT-2      | 2.5×                                   | 15.0%               | 2.0%          | 7.5%           |
| Mistral-7B | 6.6×                                   | 46.7%               | 0.0%          | 47.0%          |
| Llama-2-7B | 5.6×                                   | 47.3%               | 4.7%          | 48.7%          |

## B.8 Cross-Lingual Invariance Probe

Nothing in the construction is paraphrase-specific: it needs a grouping of contexts asserted to be equivalent and their activations. As a transfer check we use English–French translation pairs from OPUS-100 as sheaf edges on Llama-3-8B (layer 21, edge dimension 128,  $k = 20$ , seed 42), fit  $H^0$  on 100 pairs, and evaluate top-1 translation retrieval on 100 disjoint held-out pairs (Table 15). The  $H^0$  kernel retains cross-lingual retrieval and matches or exceeds the full 4096-dimensional hidden state in the EN→FR direction, while the  $H^1$  complement discards it. The decomposition separates translation-invariant content from language-specific variation as it does for paraphrase; a full cross-lingual study is future work.

Table 15: Held-out cross-lingual retrieval on Llama-3-8B (top-1 %, 100 test pairs,  $H^0$  fit on 100 disjoint training pairs).

| Subspace                     | FR→EN | EN→FR |
|------------------------------|-------|-------|
| $H^0$ kernel (20-dim)        | 76    | 78    |
| $H^1$ complement             | 6     | 3     |
| Full hidden state (4096-dim) | 82    | 59    |

## B.9 Per-Architecture Layer Profiles

The three-depth runs (Table 3, protocol *b*) measure the  $H^1$ -to-control ratio at  $L/4$ ,  $L/2$  and  $3L/4$  for all nine architectures (Figure 5). The peak depth varies by architecture:  $L/4$  for GPT-2, Mistral-7B, Llama-3-8B, Llama-2-7B, Llama-2-13B, Phi-2 and Qwen-2.5-7B, and  $L/2$  for OLMo-2-7B and Gemma-2-9B. Three depths per model cannot resolve a layer-wise shape. For GPT-2, Llama-2-7B, Llama-3-8B and Llama-2-13B, the table reports the separate 1000-sentence run at  $L/2$  (protocol *a*).

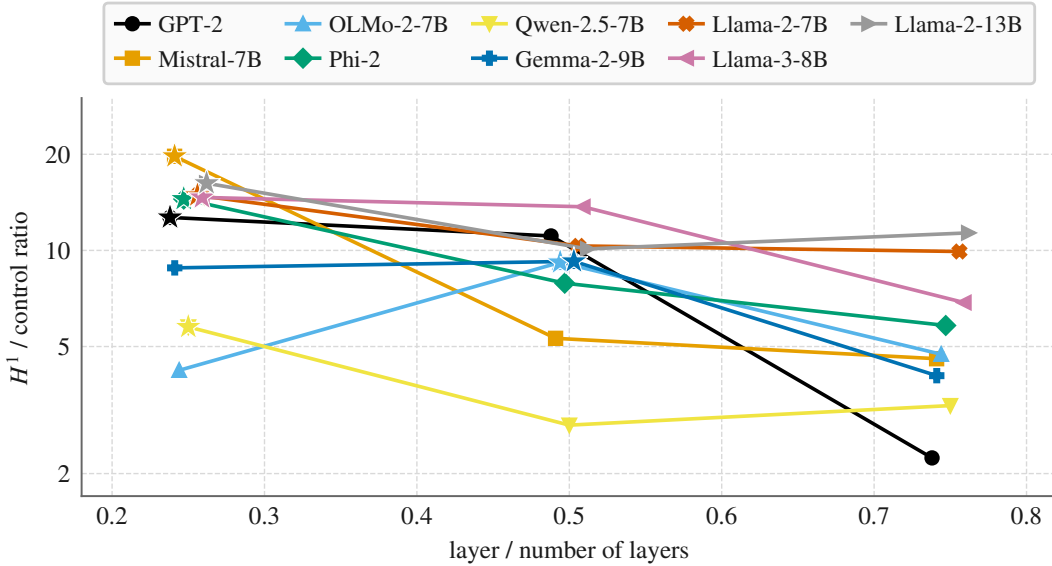


Figure 5:  $H^1$ -to-control effect ratio (protocol *b* of Table 3) at three depths per architecture,  $L/4$ ,  $L/2$  and  $3L/4$ , plotted against normalized depth (layer index over number of layers). Stars mark each architecture’s largest measured ratio, and points are offset slightly in depth so overlapping peaks stay visible. Only measured points are shown. The GPT-2 and Llama values here come from protocol *b* and differ from their protocol-*a* rows in Table 3.

## B.10 McNemar Tests with Multiple Comparisons Correction

All main comparisons (sheaf vs. random, sheaf vs. Fisher) are evaluated via McNemar’s test ( $n=1000$  per model) with Benjamini-Hochberg correction for 16 simultaneous tests ( $q < 0.05$ ). 13

of 16 tests survive correction, with the largest effects on Llama-2-7B (Cohen’s  $h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2} = +0.35$ ,  $p < 0.001$ ).

### B.11 PIXEL Head-to-Head on CounterFact

PIXEL [66] (Yu et al., 2025) extracts steering directions per layer using two complementary views of the prompt: tail-averaged hidden states (mean over the last  $k_{\text{tail}} = \text{clip}(\text{round}(0.10 \cdot \text{plen}), 3, 8)$  prompt tokens) and the end-token hidden state. Differences between paired positive and negative samples are stacked across the two views and the top principal direction (weighted PCA,  $r=1$ ) is taken as the unit direction  $u_\ell$  per layer. At inference time, for each (layer, prompt-position) site, a closed-form per-position magnitude is computed:

$$a = h^\top u, \quad B = \sqrt{\|h\|^2 - a^2}, \quad T = Bs/\sqrt{1 - s^2}, \\ \alpha = \text{clip}(\max(\max(0, -a/c), (T - a)/c), [\alpha_{\min}, \alpha_{\max}]),$$

where  $s = z_{\text{target}}$  is the target z-score (we use the PIXEL default  $s = 0.85$ ) and  $c$  is the original direction norm (here  $c = 1$ ).

We reimplement PIXEL’s direction extraction and per-position magnitude as published, but use a single global direction (extracted once from the casual/formal CAA pairs) rather than per-question directions, since CounterFact has no per-question gating signal analogous to PIXEL’s MC2 score on TruthfulQA. We inject only at the last  $k_{\text{tail}}$  prompt positions during the prompt forward pass. Repetition penalty (1.3), greedy decoding, 100 max-new-tokens, layer  $\ell = 20$  ( $\lfloor 5L/8 \rfloor$ ), and 4-bit quantization match Section 5.4; the only difference between PIXEL and our other methods is the direction-extraction recipe and the per-position magnitude calibration.

We run PIXEL at  $n = 1000$  with  $\alpha_{\text{clip}} = [0, 200]$ , which gives the closed form full freedom (results in Table 4). PIXEL exhibits the same architecture-dependent fragility pattern documented for our methods: it strongly improves over CAA on fragile Llama-2-7B but gets approached or beaten by random on robust Mistral-7B. This supports the architecture-diagnostic interpretation: the influence-concentration mechanism (Section 5.4) is shared across direction-extraction recipes. PIXEL’s dual-view direction extraction is a stronger choice than CAA’s mean-of-differences, but the closed-form per-position calibration does not overcome architecture fragility. Contrastive sheaf steering with the architecture-matched restriction map remains the strongest single method on Llama-2-7B (31.0%); CCA sheaf steering is strongest on Mistral-7B (41.6%); no sheaf method beats random on Llama-3-8B at  $\alpha = 0.30$ .

What we did not implement: PIXEL’s per-question soft-gating uses a labeled MC2 score on TruthfulQA, which CounterFact does not expose; we use a single global direction. PIXEL’s continuation-position injection (the “include-pred” option) requires modifying activations during incremental decoding with KV cache, which would constitute a larger surgery than the direction-vs-direction comparison we want here. PIXEL’s residual\_scale and auto-flip are tuned per benchmark in their paper; we keep both at default (1.0, no flip) and rely on sign-aligned PCA. These simplifications affect only the magnitude side of PIXEL; the direction-extraction recipe (the contribution most relevant to the head-to-head with our  $H^1$  direction) follows the original.

### B.12 Holonomy Null Calibration

We calibrate the cycle-holonomy obstruction  $\|I - H_\gamma\|_F$  against two nulls and a finite-sample stability check (full data in Table 16). All numbers use edge  $\dim k = 8, 11$  fundamental cycles, layer  $\ell = 8$  for GPT-2 and  $\ell = 16$  for the others, and 6 token samples per node.

**Shuffled-paraphrase null.** For each of 20 seeds, randomly permute which paraphrase belongs to which fact *within each relation*. Refit per-node PCA  $P_v$  and Procrustes transports  $Q_{uv}$  on the shuffled assignments. Recompute holonomy. Observed obstructions sit 0.10–0.29 above the shuffled null on all four architectures, and none of the differences is significant (one-sided  $p = 0.10$ –0.33).

**Bootstrap on tokens-per-node.** Refit  $P_v$  and  $Q_{uv}$  using  $N \in \{2, 3, 4, 5, 6\}$  token samples per node. On all four architectures the obstruction falls as  $N$  grows (from 3.1–4.4 at  $N = 2$  to 2.2–2.8 at  $N = 6$ ), so part of the observed magnitude is finite-sample noise in the Procrustes fits, which are underdetermined when  $N < k$ .

**Identity-transport sanity.** Setting every  $Q_{uv} = I$  gives  $\|I - H_\gamma\|_F = 0$  to machine precision, and on synthetic flat data, where every true holonomy is  $I$ , the pipeline recovers  $\|I - H_\gamma\|_F < 10^{-4}$ .

Table 16: Holonomy null calibration on the relation hypergraph. “Observed” = real paraphrase assignments. “Shuffled null” = mean over 20 within-relation paraphrase shuffles.  $p$  is the one-sided permutation  $p$ -value for  $P(\text{shuffled} \geq \text{observed})$ , smaller meaning observed obstruction exceeds null. Bootstrap- $N$  obstructions are mean  $\pm$  std over 11 cycles at the listed token count.

|              | Observed | Shuffled | $p$  | $N=2$ | $N=3$ | $N=4$ | $N=6$ |
|--------------|----------|----------|------|-------|-------|-------|-------|
| GPT-2 (124M) | 2.92     | 2.78     | 0.24 | 3.09  | 3.15  | 2.55  | 2.43  |
| Mistral-7B   | 2.52     | 2.42     | 0.33 | 3.20  | 3.31  | 2.64  | 2.18  |
| Qwen-2.5-7B  | 3.13     | 2.84     | 0.10 | 3.82  | 3.43  | 2.90  | 2.78  |
| Phi-2        | 3.01     | 2.76     | 0.24 | 4.37  | 3.37  | 3.28  | 2.36  |

Cycle holonomy on the relation hypergraph is therefore far from Haar-random but not distinguishable from a shuffled-paraphrase null at this sample size: the transports carry relation structure that survives reassigning paraphrases within a relation, and no architecture shows a fact-specific holonomy signal.

### B.13 Node-Space Operationalizations of Relation-Hypergraph $H^1$

The relation hypergraph (Section 4.2) yields  $\dim H^1 = 66$  algebraically and 18.4% Hodge harmonic mass. Lifting this back into node space did not yield a downstream win.

**Harmonic projection as features.** Project node activations onto the harmonic subspace  $\ker(L_1)$  pulled back to the node stalks via  $\delta^0$  adjoints. On 60-fact retrieval (Section 5.2, hard same-relation negatives), the harmonic projection retrieves at 36.3%, below the random-20 baseline (49.1% on GPT-2). The harmonic component encodes obstruction to gluing rather than fact identity; cycle-stable directions are stable across a relation, not within a fact, so the basis is mismatched to the retrieval target.

This result leaves the algebraic content intact, since the spectrum and Hodge mass in Appendix A.1 establish it independently. It bounds the immediate downstream usefulness of node-space projections of relation-hypergraph  $H^1$  for retrieval. Untested directions include cycle-aware steering, supervised features on the harmonic basis, and per-node restriction maps trained for downstream prediction.

### B.14 Sheaf-Regularized SSL Collapse Prevention (Per-Architecture)

Per-architecture results referenced in Section 7. A two-hidden-layer MLP encoder ( $d \rightarrow 256 \rightarrow 256 \rightarrow 64$ ) is trained on 300 MRPC paraphrase pairs for 1500 Adam steps under three loss regimes. Naive MSE and pure  $\text{tr}(L_{\mathcal{F}})$  collapse the encoder (effective rank 14–42/64 with pair MSE indistinguishable from zero);  $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$  holds effective rank in a tight 57–60/64 band across all seven architectures.

### B.15 MLST on STS-B (Full Table)

Multi-Layer Sheaf Consistency Training compared against frozen baseline, SimCSE, and VICReg on STS-B Spearman/Pearson. Frozen encoders, edge  $\dim k = 64$ ; trainable per-layer projections  $P_l$  and transitions  $T_{l \rightarrow l+1}$ ; 2000 Adam steps on MRPC paraphrase pairs (positive labels only). GPT-2 with layers  $\{3, 6, 9, 12\}$  and 10 seeds; Mistral-7B with layers  $\{8, 16, 24, 32\}$  and 5 seeds; same hyperparameters in both. The *nosheaf* ablation drops the  $T$ -transitions and the cycle-holonomy term, keeping only per-layer paraphrase consistency with the variance/covariance regulariser. Read-out modes: *final* =  $P_L^\top h^L$ ; *concat* =  $[P_l^\top h^l]_l$  stacked; *section* = concat together with  $T$ -transported predictions  $T_{l-1 \rightarrow l} P_{l-1}^\top h^{l-1}$ .

Paired- $t$  tests on per-seed Spearman, GPT-2 ( $n = 10$ ): MLST-nosheaf-concat – VICReg  $\Delta = +0.046$ ,  $p = 6 \times 10^{-9}$ ; MLST-section – VICReg  $\Delta = +0.035$ ,  $p = 1 \times 10^{-6}$ ; MLST-section – SimCSE  $\Delta = +0.080$ ,  $p = 2 \times 10^{-7}$ ; MLST-section – MLST-nosheaf-concat  $\Delta = -0.010$ ,  $p = 9 \times 10^{-4}$ . Mistral-7B ( $n = 5$ ): MLST-section – VICReg  $\Delta = -0.016$ ,  $p = 0.20$ ; MLST-nosheaf-concat – VICReg  $\Delta = -0.001$ ,  $p = 0.94$ ; MLST-section – SimCSE  $\Delta = +0.063$ ,  $p = 0.005$ . Cross-architecture rank-correlation across the eight trained methods is  $\rho = 0.74$ ,  $p = 0.037$ .

| Embedding           | Loss                                                      | Top-1 $\sigma^2$ frac. | Effective rank | Feature var. | Pair MSE |
|---------------------|-----------------------------------------------------------|------------------------|----------------|--------------|----------|
| GPT-2 (768D)        | Naive MSE                                                 | 0.149                  | 16.0/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.151                  | 16.6/64        | 0.008        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.109                  | <b>58.5/64</b> | <b>4991</b>  | 0.0149   |
| Mistral-7B (4096D)  | Naive MSE                                                 | 0.134                  | 34.4/64        | 0.005        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.121                  | 33.1/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.094                  | <b>58.0/64</b> | <b>8418</b>  | 0.0186   |
| Qwen-2.5-7B (3584D) | Naive MSE                                                 | 0.107                  | 35.9/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.119                  | 27.5/64        | 0.005        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.143                  | <b>58.5/64</b> | <b>6213</b>  | 0.0202   |
| Llama-2-7B (4096D)  | Naive MSE                                                 | 0.155                  | 36.1/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.115                  | 42.5/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.082                  | <b>59.5/64</b> | <b>6512</b>  | 0.0128   |
| Llama-3-8B (4096D)  | Naive MSE                                                 | 0.169                  | 22.2/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.115                  | 33.6/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.109                  | <b>58.0/64</b> | <b>6843</b>  | 0.0316   |
| OLMo-2-7B (4096D)   | Naive MSE                                                 | 0.168                  | 23.3/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.420                  | 14.3/64        | 0.005        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.184                  | <b>57.0/64</b> | <b>4127</b>  | 0.0134   |
| Gemma-2-9B (3584D)  | Naive MSE                                                 | 0.185                  | 23.4/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}})$ only                         | 0.145                  | 23.0/64        | 0.006        | 0.0000   |
|                     | $\text{tr}(L_{\mathcal{F}}) - \beta \log \det \text{Cov}$ | 0.104                  | <b>57.8/64</b> | <b>3686</b>  | 0.0133   |

Table 17: Collapse prevention per architecture: top-1 variance fraction, effective rank (of 64), feature variance and paraphrase-pair MSE under naive MSE, the Laplacian energy  $\text{tr}(L_{\mathcal{F}})$  alone, and the Laplacian energy with a log-determinant covariance term.

| Method                                | GPT-2 ( $n = 10$ )                  |                    | Mistral-7B ( $n = 5$ )              |                    |
|---------------------------------------|-------------------------------------|--------------------|-------------------------------------|--------------------|
|                                       | Spearman $\uparrow$                 | Pearson $\uparrow$ | Spearman $\uparrow$                 | Pearson $\uparrow$ |
| Vanilla (no training)                 | 0.428                               | 0.312              | 0.533                               | 0.498              |
| SimCSE (final-layer, InfoNCE)         | $0.578 \pm 0.014$                   | 0.567              | $0.566 \pm 0.016$                   | 0.549              |
| VICReg (final-layer, inv + var + cov) | $0.623 \pm 0.006$                   | 0.602              | <b><math>0.645 \pm 0.025</math></b> | <b>0.635</b>       |
| MLST nosheaf (final)                  | $0.621 \pm 0.005$                   | 0.598              | $0.634 \pm 0.025$                   | 0.626              |
| <b>MLST nosheaf (concat)</b>          | <b><math>0.668 \pm 0.005</math></b> | <b>0.650</b>       | $0.644 \pm 0.009$                   | 0.630              |
| MLST full (final)                     | $0.620 \pm 0.006$                   | 0.597              | $0.622 \pm 0.062$                   | 0.616              |
| MLST full (concat)                    | $0.658 \pm 0.007$                   | 0.635              | $0.631 \pm 0.033$                   | 0.618              |
| MLST full (section)                   | $0.658 \pm 0.006$                   | 0.634              | $0.629 \pm 0.021$                   | 0.611              |

Table 18: Multi-layer sheaf consistency training (MLST) on STS-B with frozen encoders: Spearman and Pearson correlation, mean  $\pm$  s.d. over seeds. *nosheaf* drops the transitions and the cycle term.

The robust positive finding on both architectures is multi-layer projection + variance/covariance regularization  $>$  single-layer baselines; the specific sheaf transition and cycle apparatus is neutral-to-slightly-negative as a training signal at this scale. The cycle-holonomy term is too easily satisfied by paraphrase-mean alignment and overlaps with the consistency loss; the  $Q$ -transitions add no consistent gradient signal beyond what concat already provides. Larger encoders, harder tasks, or richer cycle structure could change this; we leave that to future work.

## C Reproducibility

### C.1 Hyperparameter Sensitivity

The spectral gap (ratio of maximum to minimum eigenvalues) stays in  $91.7\text{--}95.6\times$  across edge dimensions  $k \in \{32, 64, 128\}$  (the count  $m$  selects eigenvectors and does not change the spectrum); the content/context boundary is stable under these choices.

**$m$  sensitivity for  $H^0$  retrieval.** We sweep  $m \in \{5, 10, 20, 40, 80, 128\}$  (with edge dimension  $k = 128$  fixed) on hard CounterFact retrieval (Table 19). The plateau spans  $m \in [20, 80]$  on all four tested architectures;  $m = 40$  is consistently slightly better than  $m = 20$  (+4 to +6 pp). The boundary case  $m = 128 = k$  leaves no eigenvectors above the spectral gap (no kernel selection) and collapses to 36–53%. The full hidden state without projection lands at 43–78%. We chose  $m = 20$  *a priori* to leave a comfortable margin from the  $k$  boundary; results are not test-tuned and are stable across the plateau.

Table 19:  $H^0$  subspace dimension  $m$  sweep (hard CounterFact retrieval, edge dim  $k = 128$ ). Results are stable across  $m \in \{20, 40, 80\}$ . At  $m = 128 = k$  no kernel is selected and retrieval collapses.

| Model      | $m=5$ | $m=10$ | $m=20$ | $m=40$      | $m=80$ | $m=128$ | Full |
|------------|-------|--------|--------|-------------|--------|---------|------|
| GPT-2      | 63.7  | 74.3   | 85.3   | <b>91.2</b> | 87.8   | 35.9    | 43.1 |
| Phi-2      | 67.3  | 75.8   | 87.1   | <b>90.3</b> | 85.3   | 52.8    | 69.1 |
| Mistral-7B | 70.9  | 82.4   | 87.1   | <b>90.3</b> | 86.7   | 48.5    | 77.7 |
| Llama-2-7B | 62.8  | 74.0   | 80.4   | <b>85.3</b> | 85.1   | 35.2    | 68.2 |

**Pooling sensitivity.** We compare three pooling choices (mean over all tokens, last non-pad token, first token) by extracting activations under each and rerunning the full  $H^0$  pipeline (Table 20). Mean pooling is best across all four architectures ( $H^0$  at 80–87%); last-token pooling drops 10–30 pp; first-token pooling collapses (autoregressive models have very limited content at position 0). Sheaf  $H^0$  beats the matched Full hidden state under mean and last-token pooling on every architecture, and under first-token pooling on three of four (on Mistral-7B, 14.2% against 20.5%). Absolute retrieval rates depend on whether the chosen pooling captures content.

Table 20: Pooling sensitivity (hard CounterFact retrieval). Mean pooling is best for all architectures, and sheaf  $H^0$  beats the full hidden state under every pooling choice except first-token pooling on Mistral-7B.

| Model      | Mean (default) |             | Last token |       | First token |       |
|------------|----------------|-------------|------------|-------|-------------|-------|
|            | Full           | $H^0$       | Full       | $H^0$ | Full        | $H^0$ |
| GPT-2      | 43.1           | <b>85.3</b> | 25.5       | 74.3  | 41.8        | 42.7  |
| Phi-2      | 69.1           | <b>87.1</b> | 26.9       | 65.9  | 30.5        | 43.8  |
| Mistral-7B | 77.7           | <b>87.1</b> | 40.4       | 72.2  | 20.5        | 14.2  |
| Llama-2-7B | 68.2           | <b>80.4</b> | 21.4       | 47.0  | 19.6        | 23.0  |

## C.2 Experimental Details

**Compute:** All experiments on NVIDIA A100 or RTX 4090 GPUs with 80GB or 24GB VRAM respectively. 7B+ models use 4-bit quantization (bitsandbytes). Sheaf construction (restriction map + Laplacian) takes < 1 minute per model; dominant cost is activation extraction (identical to any activation-based method).

**Hyperparameters:** Edge dimension  $k = 128$  (capturing  $\geq 98\%$  of variance). Mean pooling over sequence positions. Restriction map learned via joint PCA. Top/bottom 20 eigenvectors define  $H^0/H^1$ . Steering vectors norm-matched to 30% of mean hidden state norm.

**Layer selection:** retrieval and the sheaf constructions use  $\ell = \lfloor 0.67L \rfloor$  (layer 8 for GPT-2, layer 21 for 32-layer models), and steering uses  $\ell = \lfloor 5L/8 \rfloor$  (layer 7 for GPT-2, layer 20 for 32-layer models). Targets middle-to-late layers where semantic content is well-formed but output projections have not dominated.

**Statistical tests:** Bootstrap confidence intervals (1000 resamples). Mann-Whitney U for group comparisons. McNemar for paired binary outcomes. Cohen’s  $d$  (continuous) and Cohen’s  $h$  (proportions) for effect sizes. All  $p$ -values are two-sided except the Spearman and holonomy permutation tests, which are exact one-sided in the pre-specified direction.

### C.3 Practitioner Guidance

Recommended workflow: (1) Build the sheaf with 200+ MRPC paraphrase pairs using PCA restriction maps (parameter-free default). (2) Apply steering at moderate  $\alpha$  (0.10–0.15) using both  $H^1$  dims and  $H^0$ -removed, selecting the best on a validation set ( $n=50$ ). (3) For fragile models (identifiable via a single random-steering probe), use sheaf  $H^1$  to avoid generation collapse. For robust models, expect random steering to be competitive at  $\alpha \geq 0.30$ ; prefer moderate magnitudes for sheaf advantage.

### C.4 NeurIPS Paper Checklist

#### 1. Claims

**Question:** Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

**Answer:** [Yes]

**Justification:** Section 1 enumerates three contributions, with the SSL result reported in Section 7: (i)  $H^0$  beats LEACE on held-out CounterFact, the kernel selection carries the result under random or identity  $P$ , and joint multi-paraphrase constraints add +6.5 pp (Tables 2, 11, 10); (ii)  $H^1$  functional influence ( $5.6\text{--}26.5\times$  over coordinates of matched per-coordinate variability; Table 3); (iii) multi-layer grids carry a harmonic mass that ranks six architectures by steering fragility at  $\rho = 0.90$  (Table 1, Appendix A.3). Each claim is supported by experiments in Section 5 with effect sizes,  $p$ -values, and bootstrap CIs. The negative result on MLST cycle/transition terms is reported in Appendix B.15 (Table 18).

#### 2. Limitations

**Question:** Does the paper discuss the limitations of the work performed by the authors?

**Answer:** [Yes]

**Justification:** Section 7 contains an explicit Limitations paragraph covering: (a) restriction-map selection is architecture-dependent (contrastive Fisher wins on Llama-2-7B, CCA on Mistral-7B); (b) restriction-map selection is partially domain-dependent (cross-corpus  $P$  costs  $\sim 25$  pp, Appendix B.1); (c) sheaf steering wins at  $\alpha \in [0.10, 0.15]$  but ties random at  $\alpha = 0.30$  on robust models; (d) a node-space operationalization of relation-hypergraph  $H^1$  did not transfer to retrieval (Appendix B.13); (e) cycle holonomy does not differ significantly from a shuffled-paraphrase null (Appendix B.12); (f) the MLST cycle/transition terms are neutral-to-slightly-negative as a training signal (Appendix B.15); (g) our PIXEL comparison uses a simplified port (single global direction; Appendix B.11); we do not empirically compare to Angular Steering or MSRS (orthogonal frameworks).

#### 3. Theory, Assumptions and Proofs

**Question:** For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

**Answer:** [Yes]

**Justification:** The paper has one Proposition (Proposition 3.1: reduction of the sheaf Laplacian to projected within-pair covariance). The assumptions are stated explicitly in the proposition body (shared-projection matching-graph regime, one  $P$  for every node,  $Q_{uv} = I$  on edges); the failure regime under per-node  $P_v$  and Procrustes transports is also stated. The full proof is provided in Appendix A.5. Remark 3.3 clarifies the algebraic-vs-spectral  $H^1$  distinction so the cohomological language is unambiguous.

#### 4. Experimental Result Reproducibility

**Question:** Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

**Answer:** [Yes]

**Justification:** The pipeline is specified in equation form in Section 3 (joint-PCA restriction map, sheaf Laplacian as projected within-pair covariance,  $H^0/H^1$  as smallest/largest eigenspaces, mean-difference steering with  $H^0$ -removal). All hyperparameters (edge dimension  $k = 128$ ,  $H^0/H^1$  count  $m = 20$ , layer  $\ell \approx 2L/3$  ( $\lfloor 5L/8 \rfloor$  for steering), steering norm  $\alpha = 0.3\|\bar{h}\|$ , MLP encoder geometry  $d \rightarrow 256 \rightarrow 256 \rightarrow 64$ , training step counts) are documented in Section 5 and Appendix C.2. Datasets are public (MRPC, PAWS, QQP, STS-B, CounterFact) with standard splits and 70/30 train/test for the LEACE comparison. Quantization (4-bit bitsandbytes for 7B+ models) and layer selection rules are explicit.

## 5. Open Access to Data and Code

**Question:** Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

**Answer:** [Yes]

**Justification:** All datasets used are public benchmarks with standard licenses (cited in Section 5 and below). Code implementing the sheaf decomposition, steering pipeline, multi-layer grid, and MLST, with a claim-by-claim reproduction guide, is released at <https://github.com/CWRU-AISM/gluing-1c>. The construction in Section 3 together with the hyperparameters in Appendix C.2 is sufficient to reimplement the pipeline from the paper alone, and all numerical results are deterministic given the published hyperparameters and seeds.

## 6. Experimental Setting/Details

**Question:** Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer, etc.) necessary to understand the results?

**Answer:** [Yes]

**Justification:** Section 5 (Setup and Datasets) and Appendix C.2 specify: nine architectures with sizes; layer selection rule  $\ell \approx 2L/3$  ( $\lfloor 5L/8 \rfloor$  for steering); edge dimension  $k = 128$  chosen to capture  $\geq 98\%$  paired variance;  $H^0/H^1$  slice  $m = 20$ ; bootstrap CIs with 1000 resamples; McNemar’s test for paired comparisons; restriction maps (PCA / CCA / contrastive Fisher) and how each is fit; 200 MRPC pairs for sheaf construction; CounterFact  $n = 1000$  for steering; 70/30 fact split for LEACE comparison.

## 7. Experiment Statistical Significance

**Question:** Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

**Answer:** [Yes]

**Justification:** Bootstrap 95% CIs are reported alongside every steering and retrieval point estimate (Tables 3, 4, 2). McNemar’s exact test is used for paired binary outcomes; Cohen’s  $d$  for effect sizes; Mann–Whitney  $U$  for unpaired comparisons. Multiple-comparisons control via Benjamini–Hochberg is detailed in Appendix B.10. MLST results are over ten seeds with reported standard deviations and paired  $t$ -tests. All  $p$ -values are two-sided except the Spearman and holonomy permutation tests, which are exact one-sided in the pre-specified direction.

## 8. Experiments Compute Resources

**Question:** For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

**Answer:** [Yes]

**Justification:** Appendix C.2 reports hardware (NVIDIA A100 80GB or RTX 4090 24GB), 4-bit quantization for 7B+ models, and execution time (sheaf construction  $< 1$  minute per model; activation extraction is the dominant cost). Total project compute is approximately 200 GPU-hours. Section 5 indicates the cluster setup; no experiment requires more than a single A100.

## 9. Code of Ethics

**Question:** Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

**Answer:** [Yes]

**Justification:** The work is interpretability research on publicly released language models using public paraphrase and factual-recall benchmarks. No human subjects, no crowdsourcing, no scraping of private data, no release of new high-risk artifacts. Datasets used (MRPC, PAWS, QQP, STS-B, CounterFact) are standard licensed benchmarks. We have read the NeurIPS Code of Ethics and conform.

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**Question:** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

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**Justification:** The work is interpretability research aimed at improving transparency of large

language model representations. Positive impact: a label-free decomposition that exposes which directions in residual streams are content-stable across paraphrasing and which carry surface variation, potentially aiding red-teaming, alignment audits, and the diagnosis of model fragility (the steering analysis in Section 5.4 is itself a fragility diagnostic). Negative impact: the same machinery could in principle be used to construct targeted style or stance interventions against safety guardrails. We frame steering as a diagnostic tool rather than a deployment mechanism, and we report architecture-dependent failure modes (random steering can outperform sheaf steering at  $\alpha = 0.30$  on robust models, which curbs naive misuse).

#### 11. Safeguards

**Question:** Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

**Answer:** [NA]

**Justification:** The paper does not release new pretrained models, generative models, or scraped datasets. We analyze existing publicly released language models (GPT-2, Llama-2/3, Mistral, Qwen, phi-2, Gemma, OLMo) using existing public benchmarks. The released artifact is analysis code, not a model checkpoint or dataset.

#### 12. Licenses for existing assets

**Question:** Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

**Answer:** [Yes]

**Justification:** All datasets and models are cited at first use: MRPC [16], PAWS [68], QQP [36], CounterFact [47], TruthfulQA [44]. Models are accessed through their official Hugging Face releases under their respective licenses (Apache-2.0 for Mistral and Qwen; the Llama-2/3 community license; MIT for phi-2; Apache-2.0 for OLMo; Gemma terms of use; MIT for GPT-2). Each is used within the terms permitted for academic research.

#### 13. New Assets

**Question:** Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

**Answer:** [NA]

**Justification:** The paper introduces no new datasets, no new pretrained models, and no new benchmarks. All releasable artifacts are analysis code; the code repository documents them with a claim-by-claim reproduction guide.

#### 14. Crowdsourcing and Research with Human Subjects

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